

FOSTERING RESEARCH INFRASTRUCTURE ECOSYSTEMS FOR ADDRESSING COMPLEX SCIENTIFIC AND SOCIETAL CHALLENGES

OECD SCIENCE, TECHNOLOGY
AND INDUSTRY
POLICY PAPERS

September 2025 **No. 183**

OECD Science, Technology and Industry Policy Papers

This document was approved and declassified by written procedure by the Committee for Scientific and Technological Policy (CSTP) on 01/08/2025 and prepared for publication by the OECD Secretariat.

Note to Delegations:

This document is also available on O.N.E Members & Partners under the reference code
DSTI/STP/GSF(2025)5/FINAL

This document, as well as any data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

© OECD 2025



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (<https://creativecommons.org/licenses/by/4.0/>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: *In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.*

Adaptations – you must cite the original work and add the following text: *This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.*

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Acknowledgements

This work was overseen by an international Expert Group (EG, see Annex 1), nominated by GSF member countries. This draft policy report is the product of that EG's work. It was drafted by the EG Chair Roland Roberts and the OECD GSF Secretariat, Frédéric Sgard, with input from all Expert Group members. In addition, other individuals made important contributions, including experts from RIs and government institutions who participated in case study presentations (see Annex 4).

Abstract

Research Infrastructures (RIs) play a prominent role in advancing research in all scientific domains. However, individual RIs do not host all the equipment and resources required to tackle complex scientific or societal challenges, particularly when these are multi- or inter-disciplinary in nature. The development of integrated RI ecosystems offers new opportunities for enhancing the scope, impact and efficiency of individual RIs, but requires significant incentives to overcome the practical challenges that hinder their creation and sustainability. This policy report identifies and analyses good practices and presents a set of recommendations and guidance to help policymakers and RI managers support RI ecosystems across research disciplines and enhance the added value of RIs for society.

Foreword

Research Infrastructures (RIs) play a prominent role in advancing research in all scientific domains. By providing users access to a diversity of high-performance and sometimes unique scientific equipment, as well as data and digital services, RIs enable researchers to carry out projects and investigate problems that cannot be addressed at the level of a traditional small-scale research laboratory serving a limited number of users. RIs themselves are increasingly complex structures, often multimodal, and at times distributed over different sites. As data providers, they often connect a wide diversity of databases and foster relationships with local or national research institutes, post-secondary institutions, and private sector organisations.

The COVID-19 crisis demonstrated the significant potential of RIs in addressing complex societal challenges. RIs all over the world worked together in a complementary way (in RI ecosystems) to address challenges beyond their traditional scientific domains and to provide services to new categories of users, who required their products to address complex interdisciplinary questions relevant to the pandemic. While similar collaborations are pursued primarily to support excellent science, the last five years have also increasingly shown RI ecosystems' role in assisting decision-makers as they address grand societal challenges.

During the recent OECD Global Science Forum (GSF) activity on “Very Large Research Infrastructures (VLRIs): issues and options”, surveys conducted among RI managers and decision-makers identified the development of closer collaborations between VLRIs and other RIs as an interesting option for enhancing their scope and impact. Even though the development of integrated RI ecosystems holds promising potential, there are practical challenges that hinder their creation, efficiency and sustainability. Among them are issues in establishing governance models to facilitate cooperation and coordinated or integrated operations of RIs that function under very different frameworks. Funding and the sharing of material and human resources also often present a significant barrier to cooperation and collaboration, particularly across borders.

Based on its previous work on RI policy, the GSF launched a new project in 2023 entitled ‘Fostering research infrastructure ecosystems for addressing complex scientific and societal challenges’. The main output of this project, which is described in this report, is a set of recommendations and guiding tools to help policymakers and RI managers to support the formation of RI ecosystems across research disciplines and enhance the added value of these RI collaborations for society. Interesting examples of RI ecosystems and good practice examples are described in annexes.

Table of contents

Acknowledgements	3
Abstract	4
Foreword	5
Table of contents	6
Executive Summary	8
Recommendations	9
1 Introduction	11
Background	11
Methodology	12
2 Overview of the RI Ecosystem Landscape	13
What is a research infrastructure ecosystem?	13
RI ecosystems diversity	14
3 Case Study Analysis and Policy Options	17
Rationale/history	17
Main achievements	18
RI ecosystems to address complex challenges	18
Geopolitics	20
Sustainability of RI ecosystems (governance and funding)	20
User and data access	21
Human resources	22
Impact assessment	23
Guiding tools	23
4 Vision and Concluding Remarks	27
References	28
Notes	29

Annex A. Expert Group Membership	30
Annex B. Workshop agendas	32
Annex C. Survey template with examples	34
Annex D. RI ecosystems identified in landscaping	35
Annex E. RI ecosystem good practices examples	42

Figures

Figure 1. RI ecosystem composition examples	15
Figure 2. Typology of RI ecosystems based on their main strategic objectives and representative examples	16

Boxes

Box 1. Definitions	13
Box 2. RI ecosystems initiated by or involved in addressing the COVID-19 pandemic	18
Box 3. Analytical Research Infrastructures of Europe (ARIE)	19
Box 4. RI ecosystem checklist for RI managers	24
Box 5. Policy actions for enabling the development of RI ecosystems	25
Box 6. Canada's Biomanufacturing and Life Sciences Strategy: funding to the RI ecosystem	26
Box 7. Prospective views on the future of RI ecosystems	27

Executive Summary

In 2023 the Global Science Forum (GSF) mandated the OECD Secretariat and an international Expert Group with analysing the current and potential future landscape of RI Ecosystems and how they enable solutions to global scientific and societal challenges.

This report is based largely on an analysis of information collected by the Expert Group which oversaw the work (see Annex 1), composed by experts from 14 countries, the European Commission and two international organisations. It also incorporates the outcomes of two dedicated international workshops. The report provides information on the current landscape of RI collaborations and identifies and analyses good practices to develop and sustain RI ecosystems. It also identifies policies and drivers that are required to foster these collaborations and provide added value.

An exploration of existing ecosystems provided a robust picture of the current landscape and revealed a broad variety of structures with various objectives that could potentially be defined as RI ecosystems. A consensus definition for an RI ecosystem was determined for this report as a **“Dynamic collaboration between research infrastructures which has been developed around a set of shared strategic objectives and provides, supports or coordinates services and/or activities at the level of the RI collaboration, while the individual RIs remain responsible for their own governance, strategy and business”**.

The landscaping analysis showed considerable heterogeneity in the level of synergy and integration among RIs in these partnerships and a trend towards an evolution from national or regional partnerships between similar RIs to more international and multidisciplinary collaborations. It also indicated a deficit of structured ecosystems in some geographical areas and some disciplines, such as social sciences and humanities. Taking into account the diversity of RI ecosystems, the report proposes a tentative typology based on the different rationales for development of collaborative initiatives among RIs.

Detailed analyses of case studies focused on key elements of RI ecosystems, including the rationale behind their creation, their objectives and key achievements to date, including their potential for addressing complex scientific and societal challenges. This included also an analysis of: organisational, governance and financing models; user and data policies; human resource issues; and, impact assessment outcomes. Based on this analysis, a RI ecosystems checklist was developed that identifies issues that should be considered by RI managers to ensure that RI ecosystems are able to form and successfully meet their goals.

The successful development of an RI ecosystem is not solely dependent on a robust operational organisation of the partnership. Policymakers and funders also play a critical role in ensuring the conditions for sustainable and successful RI collaborations. This report discusses key policy enablers for fostering successful RI ecosystems, including supportive strategy and planning policies, dedicated financial resources and funding instruments, and adapted organisational models.

Building on the Expert Group analyses, this report lays out ten recommendations, each of which requires actions from a variety of actors. These recommendations, including the context in which they are defined, are laid out more fully in the next section of this report.

Recommendations

This section summarises ten key recommendations to advance the development of RI ecosystems. These recommendations emerged from the analysis of the survey responses of over 40 RI ecosystems (see annex 4). This survey explored the rationales for the development of partnerships between RIs, the challenges faced by RI managers in developing closer collaborations with other RIs, and the policies that encouraged their development and deepened their impact. The recommendations reflect the complexity and variety of actors and interactions that are active in a thriving RI ecosystem, and they underscore the requirement for a systemic policy approach¹. They are organised below according to specific issues, with examples of practical solutions and good practices being described in more details in the report and annex 5.

RI collaborations represent a unique opportunity for accelerating the pace of scientific discovery as well as to structure and strengthen the international research landscape (for example through enhanced collaboration between large and small/medium size RIs, and collaboration between RIs from developed and emerging economies) and increase the value and efficiency of RIs. There is need for effective information sharing to support the development of successful RI ecosystems.

Science policymakers, research funders and RI managers should recognise the unique value of RI ecosystems in supporting scientific advancement and addressing complex societal challenges. RIs should be enabled to network and add participation in RI ecosystems to their value proposition. This can be incentivised by including it as a strategic objective for individual RIs and embedding it in their mission.

Existing international fora, such as the International Conference on Research Infrastructures (ICRI) should take account of the critical role of RI ecosystems in the research landscape and provide opportunities for RI ecosystems to share good practice and lessons learned. RI road-mapping and strategic exercises play an important role in establishing funding priorities and in shaping the RI landscape.

1. **Strategic exercises and road-mapping by institutions and governments should incorporate an ecosystem view into their design objectives and therefore incentivise RI collaboration and synergies, particularly at the international level where collaboration between RIs remains challenging.**

RI ecosystem initiatives require proactive consideration of cooperation arrangements or models and may require more formalised agreements to achieve ambitious objectives. These should be flexible and responsive to facilitate the access and integration of new members, evolution of objectives and adaptation to unforeseen challenges.

2. **Governments and research agencies should explore various models of cooperation that can respond to the needs of RI ecosystems, and in particular flexibility and dynamism. There is considerable opportunity for mutual learning across countries and scientific domains in this respect.**

RI ecosystems require adequate resources to maximise the added value of synergies and collaborations. This should not be limited to financial support and is likely to be more effective when integrated into an overall strategic approach towards supporting RI ecosystems.

- 3. Research funders and governments should implement sustainable funding models for RIs to stimulate the creation of RI ecosystems and incentivise RIs to join existing ecosystems, with enhanced efforts to include the participation of RIs from emerging economies.**

Access to reliable data often remains an obstacle to the advancement of many areas of science. Although many RIs manage vast amounts of data of critical importance and have made considerable efforts to make them accessible, there is still a considerable heterogeneity in data policies at the level of individual RIs, sometimes even within the same field. RI ecosystems have the potential to act as key digital enablers within and across different research fields.

- 4. RI ecosystems should, where appropriate, include in their objectives the task of developing shared digital platforms and data access mechanisms and policies as well as standardised data formats to enhance the usefulness of the data produced by their members.**

Many RIs are facing challenges to attract and retain the highly skilled workforce they require and enhanced mobility within RI ecosystems could both improve efficiency and provide better career opportunities for RI staff. At the same time, RIs serve increasingly diverse community of users, which require training, and are expected also to transfer skills and know-how to the economy and society, which are challenging tasks for individual facilities.

- 5. RIs should be incentivised, including with funding, to leverage their capacities in training, staff development and mobility, and skills exchange with different sectors through RI ecosystems.**

While individual RIs increasingly carry out impact assessment for their activities, in response to funder and host requirements, the value of collaborative undertakings is often poorly assessed.

- 6. RI ecosystems should lead and actively facilitate activities to monitor and evaluate not just the impact of their member facilities, but also that of the ecosystems themselves. They should develop common approaches to impact assessment and building evidence to support funding applications for individual facilities and for the ecosystem.**

Research security challenges are an increasing concern worldwide. RIs play a unique role as gatekeepers to technologies and capabilities and rely heavily on international collaboration. The security challenges they face can be more acute, and more damaging, than those faced by most research and higher-education institutions. Such challenges may be better addressed collectively than by individual RIs in isolation.

- 7. RI ecosystems should develop common actions in security risk assessment and risk mitigation through the sharing of good practice, raising awareness and developing shared principles and protocols.**

The emergence of integrated RI partnerships that enhance synergies between RIs offers valuable opportunities for accelerating scientific discoveries and addressing complex societal challenges but realising these will require strategic vision and revised financial and policy incentives. Dialogue should be established between policymakers, funders, and RI managers to align strategies and support mechanisms and support RI ecosystems.

1 Introduction

Background

Research Infrastructures (RIs) play a prominent role in advancing research in all scientific domains. By providing users access to a diversity of high-performance and sometimes unique scientific equipment as well as data and digital services, they enable researchers to carry out projects and investigate problems that cannot be addressed at the level of a traditional small-scale research laboratory serving a limited number of users. RIs themselves are increasingly complex structures, often multimodal, and at times distributed over different sites. As data providers, they often connect a large diversity of databases, and foster relationships with local or national research institutes, poste-secondary institutions, and private sector organisations. Nevertheless, most existing RIs are domain-specific and often serve a restricted expert user community, an issue that was underlined in the 2020 report on “Optimising the Operation and Use of National Research Infrastructures” (OECD/Science Europe, 2020^[1]). Many RIs have unexploited potential to expand their user communities. In addition, expectations from funders and policy-makers have grown and now include an emphasis on socio-economic impact, which goes beyond the production of new knowledge (see the 2019 GSF report on assessing the scientific and socio-economic impact of research infrastructures (OECD, 2019^[2])).

Recent crises have triggered turbulence, instability, and insecurity in contemporary societies, with impacts on economies, politics, and global affairs. Science, technology and innovation (STI) are uniquely positioned to help drive the necessary transformation towards more resilient economies and societies. However, STI systems may need to reform to fulfil this potential. To support a reappraisal of STI policy, the OECD developed the Agenda for Transformative Science, Technology and Innovation Policies (OECD, 2024^[3]) which was endorsed by a 2024 ministerial declaration on Transformative Science, Technology and Innovation Policies for a Sustainable and Inclusive Future². Research and technology infrastructures are expected to play a major role in this transformation agenda as an important vector for the new STI-based solutions needed to address emerging societal challenges. Indeed, the COVID-19 crisis demonstrated the potential of RIs to address societal challenges: not only were RIs able to mobilise quickly but they adapted their activities to respond to the challenges raised by the pandemic. Thus, not only were biomedical RIs fully refocused on COVID-19 issues, but many RIs from other domains also reoriented their activities to support scientists involved in COVID-19 research (OECD, 2023^[4]). For example, physics RIs proposed their equipment for identify drug targets and therapies or their high-capacity computing facilities for the development of epidemiological models and data mining. The pandemic also fostered new coordination mechanisms and partnerships between RIs: Genomic centres and high-performance sequencing platforms linked up to facilitate the survey of variants of the virus, and in Europe, a common Alliance of Medical Research Infrastructures was established to facilitate broad user access to facilities, data and resources.

Collaboration between RIs did not start with the COVID-19 pandemic. In a number of scientific domains, RIs have long been operating in partnerships to connect complementary equipment and modalities and enable more complex experiments. IceCube, the neutrino observatory, for example, has formed collaborations on topics of common interest with other neutrino telescopes deployed in the Mediterranean and in Lake Baikal through cooperative agreements. There are also strong collaborations among fusion facilities, between synchrotrons, or research vessels for example. However, what the COVID-19 crisis demonstrated is the capacity of RIs to work together in a complementary way to address complex

challenges beyond their traditional scientific domain and to open up to new categories of users. Such synergistic collaboration is the driver of new partnerships such as the Global Ecosystem Research Infrastructure (GERI), an integrated network of site-based research infrastructures dedicated to better understand how key ecosystems across global biomes function and evolve. This, and similar such collaborations, are designed not only to support excellent science but also to inform decision-makers, addressing grand societal challenges.

Several of the examples referred to above were identified during the recently completed OECD-GSF activity on “Very Large Research Infrastructures: issues and options” (OECD, 2023^[5]). Surveys conducted for this activity among RI managers and decision-makers identified the development of closer collaborations between VLRIs – and other RIs - as an interesting option for enhancing their scope and impact. However, there are a number of practical challenges to achieving this. These include issues such as collaborative governance and coordinated or integrated operation of RIs that currently function under very different models. Funding and resource issues can also be significant barriers to cooperation particularly across borders.

Considering these issues, OECD Global Science Forum launched, in 2023, a new activity on Research Infrastructure ecosystems. The objective was to investigate the opportunities and challenges as well as the practical models for the development of integrated RI ecosystems.

Methodology

This activity was undertaken by an international Expert Group (EG) nominated by the OECD Global Science Forum (Annex 1). The EG was initially co-chaired by Maria Faury (CEA, France), and Roland Roberts (NSF, the United States), and then by Roland Roberts alone following a change in Maria Faury’s position. Experts from international organisations interested in RI policy were also invited. The Terms of Reference of the work were proposed by a Scoping Group and approved by the Global Science Forum in October 2023. Preliminary fact-finding and discussion with the EG were conducted to identify existing RI ecosystems worldwide and agree on a set of common definitions, as well as to identify existing challenges to provide a basis for surveys. A landscaping exercise was then carried out by the EG to review the diversity of existing and planned RI networks, consortia, agreements and collaborations and to identify the diversity of ecosystem models, their organisation and main objectives. This exercise was also useful in defining what is an RI ecosystem and what it is not (see next chapter).

Based on the results of this landscaping activity, a survey was then carried out among RI managers involved in RI ecosystems to provide detailed information on their rationale, history and strategic objectives, their main achievements to date, their organisational and financial models and their sustainability, their user and data access policies, their human resource policies, and their impact assessment. It is important to note that the survey, carried out on over 30 RI ecosystems, was designed to be illustrative rather than statistically representative and was focussed on analysing various options and deliverables from a diversity of examples. Survey responses do not necessarily reflect the full range of RI ecosystems but were deemed sufficiently diverse to capture both heterogeneities and commonalities.

Two international workshops specific to this activity were organised, one held virtually on 28 February 2024, and another held in hybrid format on 5 November 2024 in Amsterdam, The Netherlands (see agenda in Annex 2). Both workshops included case study presentations from a subset/sample of RI ecosystems stakeholders illustrating the challenges they face and providing interesting and practical solutions and options. The first workshop discussed issues related to rationale, sustainability and user/data access, while the second investigated more broadly the impact of RI ecosystems on the research system itself, including the perspectives of funders and policymakers. Stakeholder inputs from the survey and workshops on practices and policies provided more detailed information on key challenges and good practices, which are included in the report.

2 Overview of the RI Ecosystem Landscape

This chapter introduces the definition of RI ecosystems that is adopted in this report and its relationship and extension to other related policy terms. It also contains a brief review of the existing RI networks, consortia, agreements and collaborations, describing the current diversity of ecosystem models, their organisation and objectives.

What is a research infrastructure ecosystem?

There is no generic definition for what constitutes a RI ecosystem. It can be defined under its local or regional dimension or included within a whole international network that connect RIs to other institutions. It may include only relationships with other research institutions or connections between RIs of different sizes and domains, including all their socio-economic activities and links with industrial partners, suppliers, or policy communities.

In this report, a RI ecosystem is defined as “A dynamic collaboration between research infrastructures which has been developed around a set of shared strategic objectives and provides, supports or coordinates services and/or activities at the level of the RI collaboration, while the individual RIs remain responsible for their own governance, strategy and ‘business’”. Importantly, this definition emphasises on the added value brought by the active partnership. It excludes loose networks that have no clear common objectives and/or have no common deliverables as well as collaboratives that are centered around single RIs. Other relevant definitions for this report are detailed in Box 1.

Box 1. Definitions

Business model

Abstract representation of the RI ecosystem, which includes primarily its financial arrangement (excluding that of its members) and all core interrelated architectural and co-operational arrangements designed and developed by the ecosystem (presently and in the future), as well as core products or services the ecosystem offers or will offer to its members and users to achieve its strategic goals.

Governance

Governance encompasses structured agreements between its members to achieve objectives beyond simple coordination.

Research infrastructure

An organisational structure dedicated to providing experimental equipment, data or services for basic or applied research, enabling researchers to carry out projects and investigate problems that cannot be addressed at the level of a traditional small-scale research laboratory serving a limited number of users,

Research infrastructure ecosystem

A dynamic and evolving collaboration between research infrastructures which has been developed around a set of shared strategic objectives and provides, supports or coordinates services and/or activities at the level of the RI collaboration, while the individual RIs remain responsible for their own governance, strategy and business. RI ecosystems may have long-term lifetime or be more transient (when project-related).

Stakeholder

An individual, group, or organisation that may affect, be affected by, or perceive itself to be affected by a decision, activity, or outcome of an RI ecosystem, or portfolio. Owners and sponsors are included among the stakeholders.

Sustainability

The capacity for an RI ecosystem to remain operative, effective and competitive over its lifetime.

Source: Authors' elaboration

RI ecosystems diversity

An initial investigation was carried out to explore existing ecosystems in their diversity to help identify existing gaps and get a more comprehensive picture of the RI ecosystems landscape, that could serve as a starting point for a more detailed survey on interesting cases (the objective was not to carry out a complete and systematic inventory). This initial investigation included a broad variety of structures with various objectives that could potentially be defined as RI ecosystems (see Definitions in previous section), including partnerships that may not be fully formalised and still have a loose organisation but that have developed synergies between members. Both emerging and consolidated ecosystems examples were included in order to learn from a diversity of cases and understand challenges as well as to identify solutions. Priority was given, whenever possible, to ecosystems that include in their role a capacity to respond to societal challenges.

A generic template was agreed upon for collecting information (presented in Annex 3). During this landscaping exercise, the EG collected information regarding over 40 RI ecosystems of national, regional and international dimension, which are listed in Annex 4 together with some of their key characteristics. What emerged is a very large diversity of initiatives, that informed a number of interesting findings:

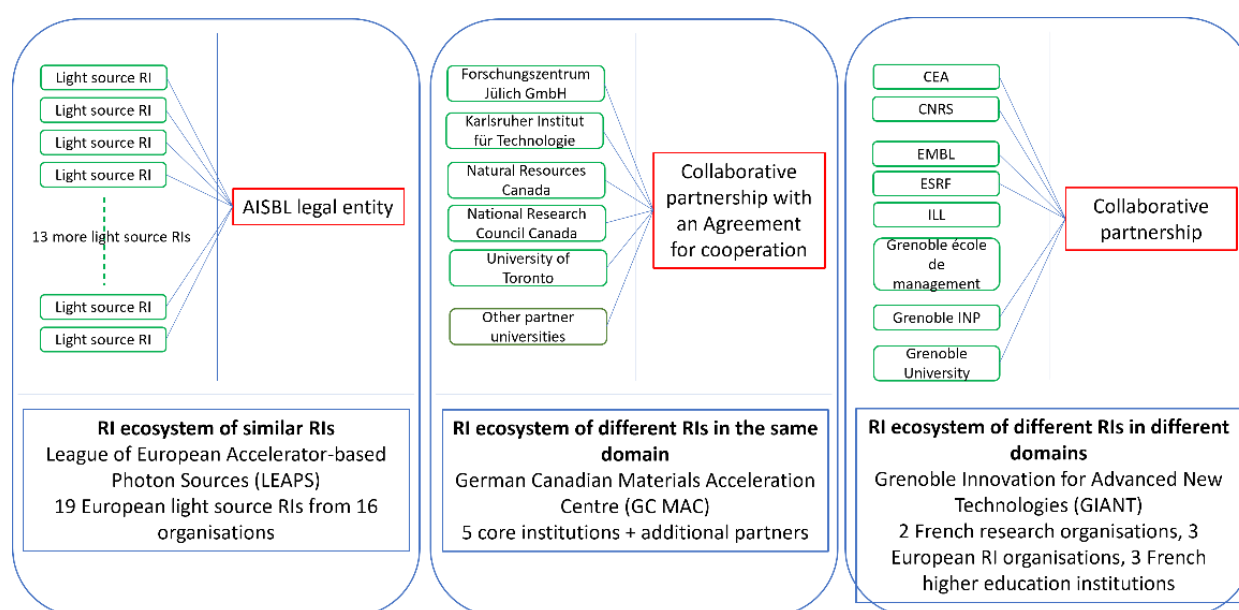
- The level of synergy and integration between RIs in these partnerships varies a lot. This landscaping survey helped determine what could be identified as a RI ecosystem (see previous sub chapter) and what should rather just be considered as a research network.
- While there is an increasing trend in the development of RI collaborative networks or consortia at national or regional level, few have an international dimension, which highlights real challenges for RIs to collaborate at international level.
- The majority of identified ecosystems were developed between similar RIs or within specific fields; the development of ecosystems that include RIs from different domains is usually more recent, suggesting that incentives for multidisciplinary partnerships have mostly been developed recently.
- RI ecosystems have a broad diversity of strategic goals. This usually includes some generic objectives, such as increasing the visibility of their user community and strengthening their interaction with funders/policymakers, but most RI ecosystems have been developed mainly on

the basis of a strong and more specific rationale - linked to overcoming shared challenges or achieving a shared ambition.

- The majority of ecosystems have been developed as bottom-up initiatives from the RI community but there are examples of top-down initiatives as a result of policy push, usually linked to dedicated funding schemes.
- The majority of the identified RI ecosystems are situated/originate from Europe and North America. There are fewer examples in Africa, Latin America or Asia-Pacific, for different reasons:
 - RIs are still scarce in Africa and have very limited resources for collaboration. The exception to this is in South Africa where RIs play a motor role across the continent.
 - The RI landscape in Latin America is evolving quickly and is still in a consolidation process. Incentives for networking are still developing.
 - Ecosystems in Asia-Pacific are often of national dimension and usually within a single scientific domain. However, regional collaborative programmes are on the rise, as there is for example a strong push from the Association of Southeast Asian Nations (ASEAN) for increased regional collaboration as well as bottom-up initiatives from specific RI communities (e.g. synchrotron facilities), and this may lead to new and strengthened RI ecosystems in coming years.
- Few ecosystems involve social sciences and humanities (SSHs) yet. Those that exist mostly regroup only RIs in the SSH domains and are not always really synergistic, despite the need for SSHs data, expertise and processes in addressing complex challenges.

The landscaping survey highlighted the broad variety of RI ecosystems. These typically fall into two main groups according to their composition: ecosystems composed of similar RIs (e.g., Laser facilities or radio-astronomy facilities) and ecosystems that includes RIs of different types and from different disciplines (see Figure 1). There is, however, a lot of diversity within these groups. For example, ecosystems composed of similar RI types may have different objectives, or ecosystems of national dimension may have very different memberships and organisational structures.

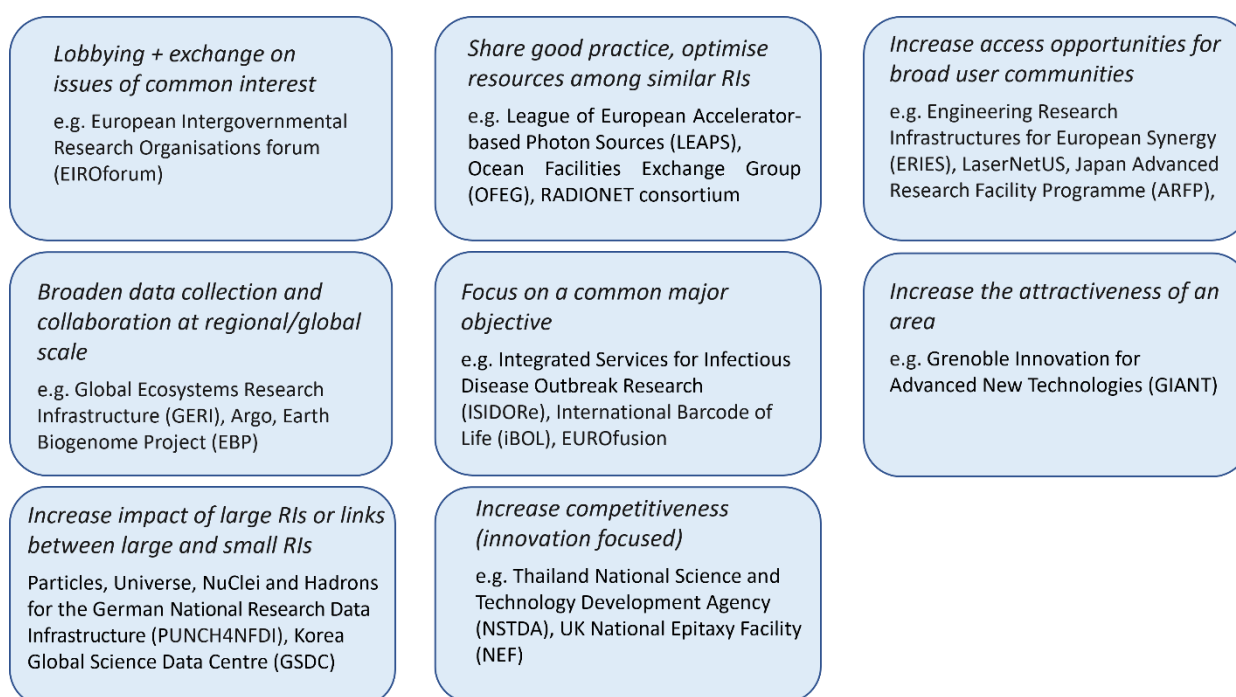
Figure 1. RI ecosystem composition examples



Note: Simplified representation of the composition of 3 representative examples of RI ecosystems
Source: OECD Secretariat

The Expert Group developed a tentative typology of RI ecosystems, presented Figure 2, based on the main rationale that has been, or is being, used to justify the development of collaborative initiatives among RIs. This typology was applied to the RI ecosystems identified during the landscaping analysis (see Annex 4) to help identify interesting cases for a more detailed analysis. Although these strategic objectives are not mutually exclusive and RI ecosystems may have been developed with several goals in mind, it was found that RI ecosystems that share key objectives often have important commonalities in their operational structures and processes.

Figure 2. Typology of RI ecosystems based on their main strategic objectives and representative examples



Note: strategic objectives are not mutually exclusive
Source: OECD secretariat

3 Case Study Analysis and Policy Options

This chapter presents a review of the rationale, objectives, organisational models and operations for RI ecosystems, as described by RI ecosystem managers when interviewed by EG members. This includes an analysis of their governance and, if relevant, financing models, as well as of their potential user and data policies, human resource aspects and impact assessment. This analysis informed the development of the guiding model, which is described in the subsequent chapter.

Rationale/history

This section articulates the motivations and reasons behind the creation of the RI ecosystems and the major steps marking their evolution since inception; For many of the RI ecosystems analysed (see the NCRIS Health group case study in Annex 5), respondents described a two-pronged rationale pertaining to the benefits of the collaboration for its RI members and for their users.

1. Expected impact on the RIs:

- Building a community and increased visibility, broadening the user community and opening up to new users;
- Increasing influence (on research funders and policy makers), more effectively leveraging existing funding, and increasing the chances of getting funding (being a member of an organised network is often perceived positively by funders);
- Promoting coordination and integration of the member RI activities;
- Supporting research on common ambitious/complex goals (beyond the capacity of single RI members). This can stem from exploiting opportunistically large top-down initiatives aiming at addressing global challenges or from bottom-up synergies in response to user;
- Coordinating or centralising RIs' proposal management or peer review systems;
- Sharing and thereby optimising human, material and capital resources;
- Sharing technologies, techniques and know-how

2. For the users:

- Increasing the chances of securing funding (applying for funds to use the capacities of an RI ecosystem is often perceived positively by funders compared to using a single RI);
- Increasing/facilitating access to RIs: enhanced visibility for users (ecosystems can better advertise their facilities), shared/simplified/harmonised access procedures ("one-stop-shop");
- Increasing/facilitating access to the data and data products originating from member RIs (in particular for international users and for industry). This can be done through single data access

portals, access to metadata resulting from compiling data from multiple RIs or training schemes for users. However, data standardisation between the multiple members of RI ecosystems remains challenging.

Main achievements

Although the impacts of RI ecosystems will depend on the time since their inception, they can usually be grouped into a small number of categories (see the LEAPS case study in Annex 5) that include;

- Structuring the relevant research domain landscape (at national or international level) by creating links between RIs and agreeing on strategic priorities, leading in some cases to the creation of new RIs or to major upgrades;
- Increasing the visibility/attractiveness of the RIs of the ecosystem to funders and users (building up a self-identity);
- Increasing the role of specific RIs into top/frontier science;
- Broadening the use of the RIs to other scientific domains/developing interdisciplinarity, including to address complex challenges (see below).

RI ecosystems to address complex challenges

The COVID-19 crisis demonstrated the added value of RIs working together to tackle complex issues (see examples in Box 2). Since then, the potential of RIs to contribute to addressing societal challenges has increasingly been put forward by decision-makers and funders, as well as by RI managers and users.

Box 2. RI ecosystems initiated by or involved in addressing the COVID-19 pandemic

The European Alliance of Medical Research Infrastructures (AMRI)

AMRI is a collaboration between three biomedical and clinical research infrastructure consortiums. The alliance aims to streamline use of scientific services, expertise, and tools and facilitate the transition of projects through different phases of the biomedical pipeline, from pre-clinical to product development. In response to the COVID-19 pandemic, the three RI consortiums partnered to establish the COVID-19 Fast Response Service to coordinate and accelerate access to facilities, services, and resources. A centralised, digital contact point is used to manage access requests and provide guidance to users on relevant services and suppliers in accelerated timelines. The service requires that users have already secured funding for their projects, although resources are available for use by applicants in preparing grant proposals.

The Collaborating Network of Networks for Evaluating COVID-19 and Therapeutic Strategies (NHLBI CONNECTS)

This collaboration was formed in the US in response to COVID-19. CONNECTS promotes rapid and efficient collaboration by integrating all major NHLBI clinical trial networks into a central adaptive platform with established master protocols. The NHLBI Biodata Catalyst provides a secure, cloud-based ecosystem for scientists to share, access, and work with digital resources and data. Critical points along the clinical spectrum, including exposure and initial symptoms to hospitalisation and recovery are targeted. Paired with the adaptive design of CONNECTS, this allows agile action to be taken to remove ineffective therapies, advance promising treatments, and adjust research as knowledge and the clinical landscape evolve. Rather than testing single treatments within single networks of clinical trial sites, researchers across sites are able to simultaneously analyse a variety of interventions, share data, and identify the most and least promising avenues of research to continue or suspend.

The Integrated Services for Infectious Disease Outbreak Research (ISIDO Re)

Developed as a lesson learned from the COVID-19 pandemic, ISIDO Re is a European initiative launched in 2022 aiming at strengthening Europe's ability to prevent, detect and respond rapidly to health emergencies. Initially funded as a three-year project by the European Commission, it is organised as a consortium regrouping over 30 bio-health and social sciences research infrastructures that provides RI services to the scientific communities through transnational access and that is developing common data platforms.

Source: Interviews with RI ecosystems managers

Grand societal and scientific challenges cut across disciplinary siloes, sectors, geographies, and government domains. As such, the development and operation of elements of science systems, and in particular RIs, deployed to address these challenges requires collaboration at the system level. Two types of RI ecosystems aiming at addressing these complex challenges co-exist: those that were specifically developed with this as their primary objective, and existing ecosystems that take advantage of the added value of working together to include this goal as a new deliverable. RI ecosystems might be permanent or transient features of the RI landscape. This flexibility/agility for assembly and dissolution, as needed, must be built into any RI ecosystem framework as it allows for models of RI ecosystem development, interaction, and evolution as the landscape of scientific and societal challenges evolves.

In addition, individual RIs are sometimes a member of more than one RI ecosystem, developing collaborations and synergies between RIs in adjacent research fields. Indeed, the evolution of cooperation between existing RI ecosystems can be important for tackling global challenges and incorporating different regional or thematic perspectives. Such dynamic systems are typically being developed in environmental and health domains, for example integrating different branches of earth sciences (e.g., in global monitoring system), or across disciplines (e.g., for studying environmental impact on human health). They may be driven by mission-oriented policies (see Box 3) or broad international agreements on societal priorities (such as the sustainable development goals adopted by the United Nations or other large UN-driven programmes). While these collaborations are often loosely organised, there are opportunities for increased synergies between RIs operating as ecosystems.

Box 3. Analytical Research Infrastructures of Europe (ARIE)

ARIE is a consortium of 7 European research networks collaborating to resolve targeted missions identified by the Horizon Europe program. The consortium has created several transversal platforms to facilitate agile and proactive advancement of fundamental, applied, and industrial multidisciplinary research from 'bench to bedside'. ARIE enabled coordination of European efforts for different aspect of the COVID pandemic response, from identification and analysis of the virus to treatment and prevention. ARIE also supports the European [ReMade](#) project designed to support the transition to a circular economy through the development of innovative, sustainable materials by providing scientists with analytical tools to explore the properties and structure of materials right down to atomic resolution. Through its delivery of services, data, and knowledge, the consortium has supported an interdisciplinary community in excess of 40,000 stakeholders from academia and industry and a diversity of scientific domains.

Source: Interview with ARIE management

Geopolitics

Geopolitics has an important influence on the broad policy context that may affect the success or the orientation of RI ecosystems. This influence is very much domain- and context-specific. For example, the COVID-19 pandemic accelerated the development of RI ecosystems (see the Japanese ARFP case study example in Annex 5). In Europe, the priorities and structure of the EC funding schemes have had a strong impact on European RI ecosystems, and, similarly, top-down policy directives can have a strong influence on the development of RI partnerships (see the GC-MAC case study example in Annex 5). International ecosystems are potentially more sensitive to geopolitical context, as they have members in different countries and often have no dedicated support funding for their joint functions.

Research security issues, which are increasingly important from a policy perspective (OECD, 2022^[6]), may have significant implications for RI operations. These may be related to cybersecurity threats, international collaboration, Intellectual Property concerns or user and staff management and mobility. Although a growing number of RIs (or their host institution(s)) are aware of the need to respond to security threats that require better risk management³, these are still poorly considered at the level of RI ecosystems. Indeed, what emerges from discussion with RI managers is that potential security threats are still largely addressed on a case-by-case basis, with limited sharing of information and good practices between RIs. RI ecosystems present a strong opportunity for sharing good practice, raising awareness and developing shared principles regarding security risks that could reduce hurdles in international collaboration.

Sustainability of RI ecosystems (governance and funding)

RI ecosystems are meta-structures that function above and beyond the governance and business model of their individual components. Ensuring the sustainability of the ecosystem was raised as a major challenge by most RI ecosystem managers, since there is usually no given organisational model or resourcing mechanism to support such initiatives. This highlights the need for a clearly defined operating framework and governance mechanisms as well as a long-term business model. A number of issues were identified, in this regard, in the survey:

- Most RI ecosystems report a lack of an adequate legal or governance model adapted to their need (particularly for partnerships with an international dimension) and have had to work around the limitations of their model to develop something practical and adapted to their objectives. Even in Europe where RIs are more used to work in networks, existing models have been described as poorly adapted to the RI ecosystems needs. For example, the European Research Infrastructure Consortium (ERIC) status in the European Union, which has been considered by a number of ecosystems as a potential solution as it has been extensively used for European distributed RIs, is also perceived as not necessarily adapted to RI ecosystems that present different characteristics and needs. The more flexible “Association Internationale Sans But Lucratif” (AISBL, International non-profit association) status has been a preferred choice for a number of initiatives, but still presents limitations for ecosystems that strive for increased synergy.
- Many RI ecosystems require an evolution of their legal model from a loose membership agreement to a more structured legal entity as they develop. This is often deemed necessary to enhance their sustainability and their capacity to build agreements with partners, to develop new and more ambitious common activities, to attract funding for shared activities, to hire staff etc. (see the case study of LaserLab EU in Annex 5). There is a demand/need for long-term legal entity/structure to achieve objectives beyond simple coordination.
- Built-in flexibility/responsiveness in the model is very important to facilitate the access/integration of new members and for members to leave as well if needed (flexibility in financial management

is also an important issue to allow a redistribution of funds to the partners in accordance with the evolution of the ecosystem), develop new objectives, respond to emerging challenges etc...

- Financial support for RI ecosystems is nearly always a challenge. In some case, part of the funding received by individual RIs may be used to coordinate with others within the collaboration, but this is rarely sufficient. For many RI ecosystems, the resources provided by members in the form of membership fees are usually enough for setting up a collaborative organisation for lobbying and for increased synergies within a scientific domain, but not for developing more ambitious common programmes or research projects. Finding the necessary resources to support the ecosystems' shared objectives is a real challenge: project-based funding streams are not sufficient and ill-adapted to support many ecosystem's objectives and there is often no adequate national or international funding mechanism appropriate for RI ecosystems. There is a need for innovative business models that enhance the sustainability of the RI ecosystems. A solution may be to develop consortia that bring together RIs and research communities/grassroot scientific associations, or mechanisms that do not necessarily involve providing financial resources (see the OFEG case study in Annex 5).
- The diversity of members in a RI ecosystem and their uneven commitments can create challenges for the development of common objectives. Similar problems can also be generated by the different timescales for which a commitment can be made by the different members and this represents an additional challenge for financial planning and ecosystem sustainability. Involving the different stakeholders (including users) in the strategy/decision-making processes is very important for the success of RI ecosystems⁴.
- Some RI ecosystems also report difficulties in the capacity of their members to support international collaboration. This is usually the case when an appropriate funding mechanism is not available in the member host country and the RI member does not have the necessary resources on its own to support such collaboration, again pointing out the need for dedicated funding mechanisms. Furthermore, setting up major initiatives/projects between RI ecosystem members often depends on external funding that may come with strings attached (there may be conflicting agenda/priorities between the individual RI funder and alternate funding sources that support and sustain the RI ecosystem), and these may also be impacted by policy changes/priorities. This pleads for the development of additional/in-house sources of revenue within the ecosystem, for example through returns on the provision of services.
- Beside the issue of financial sustainability, the survey found limited discussion yet within ecosystems over the environmental impact of RIs. A number of RI ecosystems have started to discuss the growing problem of energy cost/use (for example investigating options for shared energy procurement contracts or solutions for energy saving) but the issue of the global environmental footprint of RIs has yet to be addressed at ecosystem level.

User and data access

Facilitating and/or simplifying access to RI facilities and to the data generated by RIs is frequently indicated as a major objective for RI ecosystems, but managers also underline that this can be more complex than anticipated as each RI member often has a background of policies and use different types of languages (in data, computational, or indeed spoken language) that need to be harmonised. Furthermore, within RI ecosystems that have an international dimension, members may be subjected to specific and conflicting national rules or requirements. Survey feedback indicates that:

- There is a trend towards the development of common access portals (one-shop-stops) that can redirect users to individual ecosystem members with more or less input/contribution from the

ecosystem central hub. Similarly, there are efforts towards the development of standardised proposal submission procedures for users wishing to access the RI facilities, although proposal reviewing is still usually managed by individual RI members. There are interesting examples (see the ERIES case study in Annex 5) that demonstrate that a centralised access/proposal submission platform coupled to proposal reviewing at the ecosystem level can work.

- RI ecosystems often allow/facilitate the access of their RI member facilities to new users, but supporting access to new users requires dedicated resources (e.g., training) and specific mechanisms as those new users usually do not have the necessary expertise for leveraging the RI facilities to their full capacity.
- There is an increasing trend towards multi-disciplinary research, and therefore a demand from users to access different facility types outside their primary scientific domain. This creates both an opportunity (to access new funding streams and broaden the scope of the RIs) and a challenge (by introducing new user community needs and requiring new expertise from RI staff and reviewers) for RI ecosystems.
- There is a general trend towards FAIR (findable, accessible, interoperable, reusable) data access, as supported by many countries' overall data policies, but this is conflicting with the large heterogeneity between RI members' own data policies (or between disciplines). Overarching harmonised data management approaches, that would be suitable to all member needs, is considered as a very important objective but remains very challenging.
- There are efforts towards establishing coordinated approaches and standardisation for data access, often coupled to the production of metadata that can be accessible to larger communities of users, but this requires the harmonisation of the different RI data standards (see the ODISSEI case study in Annex 5). The obstacles are not just technical (e.g., RIs data may operate under different licensing policies) and there may be conflicting priorities between the objective of sharing data and that of preserving ownership of research or intellectual property.
- While RIs are increasingly aware of potential cybersecurity issues, this is usually managed independently by individual members of RI ecosystems. However, with a growing number of intrusion attempts reported, there is an emerging trend towards sharing of good practices among RI ecosystem members.
- There is an increasing interest for the use of AI: a number of dedicated Working Groups on this topic have been set up at ecosystem level, to explore the potential of AI to better exploit the vast amount of data produced by RIs as well as to investigate how to make different types of heterogeneous data accessible to AI/Machine Learning and to train users. Although RI managers are aware of the potential limits and ethical issues related to the use of AI, no standardised policy or guidelines for the use of AI have yet been developed at the level of RI ecosystems.

Human resources

Although RI ecosystems have various levels of integration, human resource management is usually not within their remit even in those ecosystems that have formed a legal entity. However, this is a topic that is increasingly of concern to RI ecosystem managers and for which some common issues are being discussed at ecosystem level:

- There is often no common human resource strategy within RI ecosystems (HR policies are still devolved to individual members) but the recruitment/retention of personnel is a challenge for many facilities. Some RI ecosystems are thus now working on better career paths and on gender/minority participation.

- Some RI ecosystems have developed initiatives for facilitating staff exchanges, not just for scientists but also for technical/engineering staff (see the LaserNet case study in Annex 5). RIs operate along life cycles that require various skilled personnel and it is important to have “the right people at the right place at the right time”. Mobility within the ecosystem is an essential element for increasing its efficiency. It enables the transfer of specific knowledge that was acquired and used in one RI to another, instead of losing it as missions and competencies evolve. It can also contribute to establishing clearer and more attractive career-paths in RI development and engineering.
- Staff and user training is often proposed by RI ecosystems, and the development of skilled/technical workforce is increasingly included as one of their objectives, as many of their members report difficulties in hiring and retaining their workforce.

Impact assessment

Although a few RI ecosystems have been running for several years, and the development of RI networks is often perceived favourably by funders and science policy makers alike, there is still a limited knowledge of their actual value and impact:

- The added value of RI ecosystems is rarely assessed on a systematic basis, whether internally or externally, although many ecosystems do have yearly reports that include some form of self-assessment of performance (often focused on organisation, successful funding support for their members etc.). Some are interested also in assessing their socio-economic impact (see the NEF case study in Annex 5). There are no standard frameworks or performance indicators for assessing their impact as a whole, but some RI ecosystems support the development of common impact assessment for all their members.
- Some RI ecosystems have developed support for their members in assessing the potential impact of their projects when submitting proposals to increase their chance of being funded.

Guiding tools

As introduced in the previous chapter, the concept for this chapter is to **provide guidance for RI managers on key elements to take into consideration when developing RI ecosystems and on important supporting policies that could be developed by research policy stakeholders**. Thus, in this section we provide some guiding tools to proactively foster an environment that supports the formation and success of RI ecosystems.

Based on the surveys and case studies undertaken during this activity (see Annexes 4 and 5), the RI ecosystems checklist described in Box 4 provides elements that should be considered by RI managers wishing to set up collaborative partnerships to ensure that RI ecosystems are able to form and successfully meet their goals/objectives. It should be emphasised that productive collaboration of RIs should be intentional and not left to chance. At the same time, it should be recognised that not all RIs may find it productive to enter such collaborations and that when collaborations do develop, the resulting ecosystem may be transient or develop into something that is more permanent. In the latter situation the membership arrangements should be such as to allow for the inclusion of new RIs and the exit of those already in the partnership, under clearly defined conditions.

Box 4. RI ecosystem checklist for RI managers

Rationale/strategic objectives

Agreeing on strategic priorities and common objectives is critical to the success of RI ecosystems. Future members need to clearly define the nature of the activities that will be developed under the scope of the ecosystem. This can involve extensive consultation between partner organisations and the development of a strategic plan, as well as agreeing of issues such as intellectual property rights and data policies. While agreements may evolve over time, shared priorities are required to ensure the commitment of all members and attract the necessary support from users and external stakeholders.

Geopolitics and security

RI ecosystems need to be responsive to emerging constraints generated by the evolution of the global political context. This is particularly important in relation to societal challenges and to potential security threats. RI ecosystems should serve as fora and testbeds for discussions and pilot actions to address shared challenges (such as environmental sustainability and energy/resource use) and develop shared principles regarding security risks.

Governance of the RI ecosystem

RI ecosystems require structured agreements between its members to achieve objectives beyond simple coordination, to attract funding and enhance their sustainability. There is no perfect model for RI ecosystem governance, especially when there are members from different countries. Flexibility/responsiveness is very important to facilitate the access/integration of new members, develop new objectives and respond to unforeseen challenges. RI ecosystems should establish collaborative governance mechanisms, with clear roles and accountability, that bring together relevant experts and stakeholders. Establishing a dedicated secretariat to assist with the RI ecosystem operations and develop appropriate strategies to address new challenges is also recommended.

Financing RI ecosystems

Finding the necessary resources to support the ecosystem' objectives can be a real challenge as there is often no adequate national or international incentive nor funding mechanism dedicated to RI ecosystems. There is a need for innovative business models that generate the sustained resources required for the ecosystem's operations. This can integrate diversified, funding sources such as public-private partnerships, international funding, in-kind contributions, etc.

Data governance and user access

RI operating in an ecosystem must have clear understanding/agreements related to data and user access. There should normally be an added benefit of the ecosystem for users of the individual RIs. RI ecosystems should have a shared digital and data access strategy as well as standardised proposal submission procedures that foster open science. This can encompass: piloting and experimenting with novel approaches to data standards; harmonisation of procedures for open access to data that preserve rights and research ownership; shared cybersecurity policies; and, other outputs such as standardised meta-data, methods, protocols, software, and code.

Human resources

Human resource policies adopted at the RI ecosystem level should foster a level of integration that leverages training and staff development of its individual RIs. Promoting mobility and rotation of staff across RIs should be encouraged to improve efficiency and provide better career opportunities.

Success metrics and evaluating success

As the operation of RI ecosystems can require significant resources from members, external funders and research institutions, **emphasis must be placed on clearly articulating the added value of the collaboration to all relevant stakeholders.** Impact assessments should be carried out, using appropriate performance and impact indicators, that are relevant for both RI members and RI users.

The successful development of RI ecosystems is not solely dependent on a robust operational organisation. Policy makers and funders also play a critical role in ensuring that the option to initiate or join an ecosystem is built into the terms and conditions of individual RIs. Collaborative activities at local, regional or international scales, if deemed a useful pursuit, should be costed and sufficient levels of funding should be available to support those initiatives. Additionally, policymakers shape the operating context for RIs around issues such as research security, intellectual property, or international cooperation. Potentially useful policy actions for fostering successful RI ecosystems are described in Box 5.

Box 5. Policy actions for enabling the development of RI ecosystems

The concept of RI ecosystems should be included in strategic planning exercises (RI roadmaps etc.). RI strategic exercises often constitute strong incentives and guidance for funding priorities and including RI ecosystems in these exercises would help raise the awareness of their usefulness (note that ecosystems can be transient). RI ecosystems represent excellent opportunities for North-South and South-South scientific collaboration and should therefore be integrated in regional and international strategic planning.

One of the most significant obstacles to the development of RI ecosystems is the fragmented approach to developing these collaborative partnerships within and across countries. This fragmentation stems from differences in individual research infrastructure policies (on data, intellectual property etc.), culture and programmes. **Governments and research institutions should promote and incentivise coherent RI policies (particularly for data management)** that can facilitate collaboration.

There is currently lack of adequate organisational/legal models for international RI ecosystems. **Governments and research agencies should investigate new forms of organisational models**, including legal entities where appropriate, that respond to the need of RI ecosystems as well as of other integrated international research collaborations.

Funders need to develop specific funding mechanisms that promote and recognise the value of collaboration between RIs and the development of RI ecosystems. Besides dedicated program funding for RI ecosystem development, including the participation of RIs in RI ecosystems as a merit criterion or a criterion of eligibility for funders, could constitute a strong incentive for collaboration. Incentives could also take the form of pilot projects that demonstrate the value of RI ecosystems and their challenges early on

There is a lack of specific indicators to assess the added value of RI ecosystems. Funders should support the development of indicators and methodologies that could be used in impact and performance evaluation.

Existing links between partners often constitute a prerequisite to the capacity of RI ecosystems to address complex challenges (building on existing collaboration is much easier than starting from scratch). **Funders and research institutions should therefore incentivise collaboration and synergies between RIs**, and codevelop with RIs challenge-oriented projects that are identified as strategic priorities.

Considering the complexity of the RI landscape amidst strained financial resources, there is a need for funders and decision-makers to work together and promote a positive environment for the development of efficient and sustainable RI ecosystems. Mission- and challenge-oriented funding schemes (Larrue, 2021^[7]) can provide an opportunity for multiple RIs to cooperate to better leverage and help structure existing research (and innovation) bases. One such example is Canada's recent approach to revitalising its domestic life sciences and biomanufacturing sectors (see Box 6) in the wake of the COVID-19 pandemic. Mission-oriented approaches can help mitigate gaps in the research funding landscape but must do so without generating superficial partnerships or additional layers of bureaucracy. Ideally such approaches should help connect, structure and boost existing working partnerships and networks.

Box 6. Canada's Biomanufacturing and Life Sciences Strategy: funding to the RI ecosystem

In 2020/21, the Government of Canada mobilised stakeholders across the entire life sciences domain to develop a comprehensive [Strategy](#)⁵ focused on quickly and durably mitigating gaps in the country's pandemic readiness and ability to contribute competitively on the global scene through structuring and strategic investments along the biomanufacturing value chain.

Adopting an end-to-end, whole-of-ecosystem approach with the engagement of several ministries, the *Strategy* set in motion wide-ranging and coordinated programming across academia, government and industry, including upstream investments approximating 2.2 billion CAD (notwithstanding mandatory partner funding) into foundational elements of science. As such, this represents the first-of-its-kind, large-scale "mission-oriented" RI funding initiative in Canada.

As part of these investments, the Canada Foundation for Innovation (CFI), Canada's main funder of RI, in coordination with domestic funding counterparts launched [three sequential and coordinated calls for proposals centred around RIs](#) – notably RIs for surveillance, diagnostics and therapeutics development, and high-level biocontainment facilities – but also covering components for research and talent attraction and development. The calls were designed to identify, leverage and boost existing networks of players and RI capabilities across sectors, and create meaningful synergies within the pan-Canadian RI ecosystem. Central to this aim was the *Strategy*'s provision of a robust governance structure for decision-making and oversight which provided guidance, strategic input, and feedback throughout the different stages of the merit review process. This ensured that the final recommendations of the Strategic Committee for all three calls, represented the most globally relevant and coherent portfolio of investments in alignment with the *Strategy*.

An interesting feature to this seemingly "top-down" initiative is the incorporation of a "bottom-up" element: the selection for funding of [five complementary Hubs of excellence](#)⁶ through which the top-down aims of the *Strategy* could be more efficiently realised. Each hub contributes a critical mass of RI assets and well-established linkages; proposes solid governance to stir the efforts of their coalition of partners and maintain their cohesion; and serves as a single point of entry to the vast ecosystem of academic research and its connection to industry. Once identified, the Hubs were responsible for receiving and preselecting within their coalitions, and among themselves, the best proposals according to their comparative strengths and complementarities for review by the CFI and partners. Moving forward, the Hubs were also entrusted with a central role in the implementation, oversight and accountability for the funded projects, in addition to that of the funders, as well as for developing a common set of objectives and vision to prolong their sustainability on the longer term.

Ultimately, the benefits of this pilot approach will be gauged following the monitoring activities planned in the near term as part of the *Strategy*'s oversight framework. It is nevertheless agreed upon among the many partners involved that this first experience – while challenging to coordinate – provides a good starting point for the successful delivery of other mission-driven initiatives in priority domains for a large group of stakeholders in Canada.

4 Vision and Concluding Remarks

In order to test some of the preliminary conclusions that emerged from this project, a short visioning exercise was conducted with Expert Group members and RI ecosystem managers. This included a survey on the future landscape of RI ecosystems and a summary of the survey responses is provided in Box 7.

Box 7. Prospective views on the future of RI ecosystems

Q1: What might be the most important drivers for developing RI ecosystems in coming years?

1. Facilitating scientific breakthrough/frontier research and addressing complex and multidisciplinary challenges (although the emphasis on these different drivers may differ between RIs and funders).
2. Enhancing the efficiency (governance, organisation...) of the RI system (more a priority for funders).
3. Increasing data and user access.

Q2: Where are future RI ecosystems most likely to develop in the next 10 years?

1. Regional dimension seems to be the most likely focus for new ecosystems. Europe is ahead in that respect, but there are opportunities and ongoing discussion in Latin America, Asia-Pacific and Africa for new partnerships.
2. National and International levels both offers opportunities, but there are complicated challenges for international/global collaborations. Purely national ecosystems may offer less added value. Local ecosystems could potentially generate more opportunities for technology transfer and regional economic development.

Q3: What is the most likely composition of future RI ecosystems?

1. Ecosystems composed of heterogeneous RIs in size/organisation but of the same domain are considered as most likely to be developed as members share common features and scientific goals that would facilitate partnerships.
2. Interdisciplinary ecosystems are likely to be more common. While there are challenging to develop, they offer considerable potential for responding to emerging challenges, bringing complementarities and unique capacities to the table.

Q4: What may be the most important obstacles in the future development of RI ecosystems?

1. Sustainability is perceived as the most important challenge, as dedicated resources are still rare.
2. Barriers to international collaborations (such as increased security concerns) are likely to be significant obstacle.
3. While there are important, obstacles related to practical organisation or heterogeneous RI policies and practices, these considered as manageable if there is a strong will (and resources) to pursue collaboration.

Q5: What are the key policies that should be implemented to promote the future development of RI ecosystems?

1. Including RI ecosystems in national/international RI strategies is considered as potentially a very strong incentive. In some countries, addressing the current fragmentation of the RI landscape is also considered as a prerequisite.
2. The development of dedicated funding streams is underlined as a necessary condition for thriving ecosystems
3. Recognising (and monitoring) the added value of RI ecosystems should help getting recognition and support

The key messages that surfaced during the visioning exercise reinforce the main recommendations laid out at the start of this report. Although RI ecosystems can be extremely diverse in their composition, organisation and objectives (see Annex 4), their future development is dependent on a number of common

drivers, e.g., increased efficiency, open access, research direction, emerging needs and challenges. These drivers may be perceived differently whether one takes the top-down view of policy makers or the bottom-up view of RI managers and users. Increasing the overall agility and efficiency of the system may be a top priority for funders/policy makers, while for RI managers and users, the capacity to deliver the best science will remain the primary driver. Thus, a complementary mix of top-down (roadmaps/strategic exercises, mission-oriented policies) and bottom-up actions (according to user needs) is required.

Another potential driver is the inclusion of RI ecosystems (i.e., their formation and persistence) in countries' or institutions' strategic planning exercises, or in domain-specific long term planning exercises, such as in the Canadian example above (Box 6). RI roadmaps and similar strategic planning exercises have considerable influence on RI support and policy orientation. Including RI ecosystems in these processes can enhance their visibility to relevant stakeholders and encourage the development of partnerships. This in turn can strengthen the argument for future funding and stimulate stronger engagement and commitment from policymakers (missions and challenges embedded in strategies and roadmaps provide tangible goals around which decision-makers can coalesce).

Regarding the optimal geographical scale for future RI ecosystems, there is a general agreement that regional or international partnerships have strong added-value potential, provided that appropriate funding mechanisms are developed in parallel. Similarly, the increasing push towards more interdisciplinarity (noting that even RI ecosystems that start with a disciplinary focus tend to integrate more interdisciplinary objectives) should be encouraged. However, the geographical distribution and disciplinary focus of an RI ecosystem, should be flexible and align with its intent and goals.

Finally, the need for clear goals, metrics of success, and plans for assessing such success are deemed essential and should be considered and planned for very early in the formation of RI ecosystems.

The picture that emerges from this work is that of a growing self-organisation of the RI landscape worldwide, supported both by a need for better efficiency and for rationalising resources, and for providing users with increased opportunities to combine RI services (equipment and data). The emergence of integrated RI ecosystems with enhanced synergies offers new opportunities for scientific discovery and for research addressing complex societal challenges, but requires recognition, together with financial and policy incentives. In some senses, a more professional approach is now necessary to address a number of common and recurrent practical challenges. A dialogue needs to be established between funders, policymakers, RI managers and users to align strategies and support mechanisms to support RI ecosystems.

References

- Larrue, P. (2021), "The design and implementation of mission-oriented innovation policies: A new systemic policy approach to address societal challenges", *OECD Science, Technology and Industry Policy Papers*, No. 100, OECD Publishing, Paris, <https://doi.org/10.1787/3f6c76a4-en>. [7]
- OECD (2024), "OECD Agenda for Transformative Science, Technology and Innovation Policies", *OECD Science, Technology and Industry Policy Papers*, No. 164, OECD Publishing, Paris, <https://doi.org/10.1787/ba2aaf7b-en>. [3]
- OECD (2023), "COVID-19 and policy for science", *OECD Science, Technology and Industry Policy Papers*, No. 152, OECD Publishing, Paris, <https://doi.org/10.1787/8f86e60b-en>. [4]

- OECD (2023), “Very Large Research Infrastructures: Policy issues and options”, *OECD Science, Technology and Industry Policy Papers*, No. 153, OECD Publishing, Paris, <https://doi.org/10.1787/2b93187f-en>. [5]
- OECD (2022), “Integrity and security in the global research ecosystem”, *OECD Science, Technology and Industry Policy Papers*, No. 130, OECD Publishing, Paris, <https://doi.org/10.1787/1c416f43-en>. [6]
- OECD (2019), “Reference framework for assessing the scientific and socio-economic impact of research infrastructures”, *OECD Science, Technology and Industry Policy Papers*, No. 65, OECD Publishing, Paris, <https://doi.org/10.1787/3ffee43b-en>. [2]
- OECD/Science Europe (2020), “Optimising the operation and use of national research infrastructures”, *OECD Science, Technology and Industry Policy Papers*, No. 91, OECD Publishing, Paris, <https://doi.org/10.1787/7cc876f7-en>. [1]

Notes

¹ Promoting cross-sectoral, multidisciplinary collaboration in the process of policy formulation by taking proper account of the crucial linkages between issues generally treated separately within different specialisations and scientific and institutional “silos.”

² <https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0501>

³ This topic was recently debated during a dedicated workshop on “Research security and consequences for Research Infrastructures” organised by the OECD Global Science Forum and the Commonwealth Scientific and Industrial Research Organisation (CSIRO), to explore and discuss the implications of security concerns for RI governance and operation (<https://icri2024.au/side-events/>).

⁴ The European Social Science Infrastructure White Paper (<https://zenodo.org/records/10410991>) proposes an innovative concept of building up a Research Infrastructure ecosystem that involves RIs and research communities/grassroot scientific associations. The creation of a Federation of European Social Science Infrastructures is proposed, constituted by two sets of organisations: infrastructures or service providers and field organisations. These organisations would pay a small membership fee to be part of the ecosystem and in return would be ensured that RIs serves the needs of scientists and leads to groundbreaking research.

⁵ [Overview of Canada’s Biomanufacturing and Life Sciences Strategy](#)

⁶ [Canada Biomedical Research Fund](#)

Annex A. Expert Group Membership

Country	Name	Affiliation
Australia	Tony Rothnie	Director (former), National Research Infrastructure Policy and Engagement Team, Research Policy and Programs Branch, Department of Education
	Grace Campbell	Policy officer, NRI Policy and Engagement Research Policy and Programs Branch, Department of Education
	Jason Palazzolo	Science Policy Fellow, National Research Infrastructure Policy Team Research Policy and Programs Branch Australian Government Department of Education
Canada	Heidi Bandulet	Special Advisor for Research Facilities, Canada Foundation for Innovation (CFI)
	Cherif Ould Hamou	Senior Manager with Laboratories Canada of Public Services and Procurement Canada (Government of Canada)
	Sanjeevani Kumar	Analyst, Laboratories Canada of Public Services and Procurement Canada (Government of Canada)
	Philippe-Olivier Giroux**	Manager, research infrastructure Directorate, Innovation, Science, and Economic Development Canada
	Dimitri Treheles**	Policy advisor, research infrastructure Directorate, Innovation, Science, and Economic Development Canada
Czech Republic	Jan Hrusak	Scientific advisor at the Czech Academy of Sciences
	Ondrej Hradil	Research Infrastructures Coordinator, Masaryk University
European Commission	Agnes Robin	Head of Sector Research Infrastructures, DG Research and Innovation
France	Maria Faury*	Director of international affairs and large research infrastructures, CEA
	Michel Guidal	Chairman of the Very Large-Scale Research Facilities Committee
	Eric Humler	Deputy scientific director of the National Institute for Earth Sciences and Astronomy at CNRS and chairman of the Very Large-Scale Research Facilities Committee
Germany	Renata Feldmann**	German Federal Ministry of Education and Research (BMBF), division 711 "Matter and Universe"
	Christoph Peschke	DLR Project Management Agency
	Charlotte Rothfuchs-Engels	DESY Project Management Agency
Japan	Ayano Tanabe	Specialist, Research Environment Division, Science and Technology Policy Bureau, MEXT
	Akihito Kumamoto	Senior researcher, Research Environment Division, Science and Technology Policy Bureau, MEXT
	Junna Kawaguchi	Officer, Research Environment Division, Science and Technology Policy Bureau, MEXT
Korea	Yong-Joo Kim	Lead manager, Policy team, National Research Facilities and Equipment Center (NFEC)
	Sun Kun Oh	Konkuk University
Netherlands	Patricia Vogel	Strategic policy advisor, Institutes Organisation of NWO (Dutch Research Council)

Norway	Solveig Flock	Director for the Department for Research Data and Infrastructure, Research Council of Norway
	Olja Toljagic	Senior adviser, Department for Research Data and Infrastructure, Research Council of Norway
Poland	Agnieszka Matera-Witkiewicz	Wroclaw Medical University Biobank
Switzerland	Tobias Hodel	Digital Humanities, University of Bern
Thailand	Chinorat Kobdaj	Assistant professor, School of Physics, Institute of Science, Suranaree University
United Kingdom	Beth House	Associate Director of International affairs, STFC, UKRI
United States	Roland Roberts Co-chair	Deputy Chief Officer for Research Facilities National Science Foundation
ESFRI	Jose Luis Martinez	ESFRI chair, Director General ESS-Bilbao
Science Europe	James Morris	Science Europe office
OECD	Frédéric Sgard	GSF secretariat
	Masatoshi Shimosuka	GSF secretariat
	Carthage Smith	GSF secretariat

* Maria Faury co-chaired the Expert Group until her change in responsibilities at CEA in summer 2024

** Philippe-Olivier Giroux, Dimitri Treheles and Renata Feldman were absent during the second part of the project due to changes in their responsibilities

Annex B. Workshop agendas

Workshop 1 - 28 February 2024 (via video conference)

Introduction

Presentation of the activity and workshop goals by the GSF expert group co-chairs and secretariat (Maria Faury and Roland Roberts, co-chairs of the Expert Group, and Frédéric Sgard, GSF Secretariat).

Session 1: Why should you develop RI ecosystems? Enhancing the value and impact of RIs and responding to complex challenges through synergies and partnerships

Session chair: Maria Faury

- Feedback from the GSF secretariat on the survey carried out among RI ecosystem managers
- Case study presentations:

Asia Oceania Forum for Synchrotron Radiation Research (AOFSRR), Richard Garrett, Secretary AOFSRR, Senior Advisor ANSTO

Ocean Observatory Initiative (OOI), James B Edson, Principal Investigator, Ocean Observatories Initiative, Senior Scientist at Woods Hole Oceanographic Institution

European Social Sciences and Humanities Open Cluster, Bonnie Wolff-Boenisch, chair of the EU SSH Open Cluster, Director General CESSDA

Discussion

Session 2: How to ensure the sustainability of RI ecosystems: possible governance and business models

Session chair: Cherif Ould Hamou (Public Services and Procurement Canada)

- Feedback from the GSF secretariat on the survey carried out among RI ecosystem managers
- Case study presentations:

South Africa Environmental Observation Network (SAEON), Mary-Jane Bopape, Managing Director SAEON, National Research Foundation (NRF)

Eurofusion, Tony Donné, Programme manager Eurofusion

Discussion

Session 3: How to increase access to RIs through synergistic ecosystems: User access and data policy

Session chair: Roland Roberts

- Feedback from the GSF secretariat on the survey carried out among RI ecosystem managers
- Case study presentations:

Engineering Research Infrastructures for European Synergies (ERIES), Gerard O'Reilly, ERIES Deputy coordinator, Professor at the School of Advanced Studies IUSS Pavia

ARGO, Yann-Hervé de Roeck, Director General Euro-ARGO

Discussion

Concluding remarks by chairs

Workshop 2 - 5 November 2024 (Nikhef Institute for Subatomic Physics, Science Park of Amsterdam)

Introduction (Roland Roberts, chair of the Expert Group and Frédéric Sgard, GSF secretariat)

Session 1: Sustainability-funding of RI ecosystems

What are the existing challenges in developing robust business models for sustainable RI ecosystems? What funding mechanisms could be developed to support these initiatives and ensure their long-term development? How can RI ecosystems funding streams be adapted to the ecosystems' objectives?

- Case studies from funders:
 - *European Commission* (Agnès Robin, Head of Sector Research Infrastructures),
 - *Canada Foundation for Innovation* (Heidi Bandulet, Special Advisor for Research Facilities),
 - *Dutch Research Council NWO* (Alice Dijkstra, Senior Policy Advisor)
- Panel discussion (moderator: Roland Roberts)

Session 2: RI ecosystems for addressing complex global challenges

How can RIs develop synergies and collaboration to broaden their objectives and user-base? What are the obstacles to regional/international and/or interdisciplinary collaborations between RIs? What are the potential policy incentives that could be developed by research institutions?

- Case studies from RI ecosystems and research institutions:
 - *European Plate Observing System (EPOS)* (Lilly Freda, Executive Director),
 - *Japan RIKEN Spring-8 Center* (Tetsuya Ishikawa, Director),
 - *South Africa Department of Science and Innovation* (Imraan Patel, Deputy-Director General)
- Panel discussion (moderator: Frédéric Sgard)

Session 3: Potential impact of RI ecosystems for research and the research system

How can agencies/institutions integrate RI ecosystems in their research strategy? How can RI ecosystems and research agencies/institutions co-develop ambitious strategies for increased impact in the future?

- Short summary of the brainstorming foresight session (Roland Roberts)
- Panel discussion with RI ecosystems leads and research policymakers (moderator: Patricia Vogel, NWO-I)
 - Ann Schwartz (Assistant Director for Research Infrastructure, OSTP),
 - Stéphanie Wyss (Science Europe/Swiss National Foundation),
 - Tom Emery (Deputy-Director, Open Data Infrastructure for Social Science and Economic Innovations ODISSEI),
 - Izabela Rottmann (scientific Director, RadioNet)

Conclusions/lessons learned

Annex C. Survey template with examples

RI ecosystem details	Examples
Name of the ecosystem (if any)	EUROfusion, Global Ecosystem Research Infrastructure (GERI)...
Contact for further details if needed	Web page, contact e-mail...
Main goals/objectives	Share data and provide common access to data bases, lobby funders/policymakers, develop complementarities in equipment/capacities, address complex scientific or societal questions...
Organisation structure	MoU, consortium, association...
Creation date/duration in activity	Provide information if terminated
Members/partners	RIs, research institutions, universities...
Budget (if any)	Central/common budget only, not separate budget of the ecosystem components
National/international dimension	Local, national, regional (i.e. Europe), global
Scientific domain(s)/ RI types/interdisciplinarity level	e.g. Laser RIs, environment observation RIs...
Original/unique features	Barter system as business model, centralised access mechanism...
Any other element of interest	Link to policy making, link to projects or mission, impact considerations, inclusion/diversity elements...
Comments	Impact in its domain, possible case to be further investigated in more detail.

Annex D. RI ecosystems identified in landscaping

Ecosystems	Experts Interviewed	Geographical dimension	RI types/disciplines	Drivers/Date of creation	Unique features	Typology/rationale
Asia-Oceania Neutron Scattering Association (AONSA)		Asia + Oceania	Neutron facilities	Enhance regional utilisation of neutron sources/2011	Asia-Oceania Neutron School for Young Research Fellows	Share good practice, optimise resources among similar RIs + Increase access opportunities
Asia-Oceania Forum for Synchrotron Radiation Research (AOFSTR)	Richard Garrett	Asia + Oceania	Synchrotron facilities	Bottom -up/2006	Promote the use of synchrotrons in developing economies of the region, new focus on societal challenges	Share good practice, optimise resources among similar RIs + Increase access opportunities
Advanced Research Facilities Platform Program (ARFP)	Takanori Kigawa	Japan	NMR , MRI , Laser , Microscope imaging	Top-down ministerial push/2021 (5y project)	Set up during COVID pandemic to facilitate access	Increase access opportunities for broad user communities
ARGO	Yann-Hervé de Roeck	International (20 countries+)	Ocean environment observation	Top-down international push + bottom-up international community/1999	Ground-breaking data management	Broaden data collection and collaboration at global scale
Analytical Research Infrastructures of Europe (ARIE)	John Collier	Europe	Physics RIs (Neutron sources, Electron sources, Photon sources, Lasers, Ion sources, Proton sources and High	Top-down (EC call) + bottom-up consortium of analytical research infrastructure networks grouped in a consortium for multidisciplinary uses	Provides free access to the scientific user community; multidisciplinary	Focus on a common major objective

36 | FOSTERING RESEARCH INFRASTRUCTURE ECOSYSTEMS

			Magnetic Fields)	and respond to societal challenges/2019		
Australian National Collaborative Research Infrastructure Health Group (NCRIS Health Group)	Merran Smith and Ian Smith	Australia	Bioplatforms Australia National Imaging Facility Phenomix Australia Population Health Research Network Therapeutic Innovation Australia	Bottom-up/2022	Provides a 'one stop shop' for researchers who need access to health and medical RI. Co-ordinate collaborative funding applications to granting programs to maximise efficiency of public investment.	Increase access opportunities for broad user communities
Biological and Chemical Oceanography Data Management Office (BCO-DMO)		United States	Oceanographic ecosystem data	Bottom-up + NSF driver (NSF award)/2006	Active in technology research and policy/best-practices development within data/information management	Increase access opportunities for broad user communities
Canada Biogenome project	Steven Jones	Canada	Genome sequencing and biodiversity	Bottom-up/2021 (5y project)	Part of the international Earth BioGenome Project	Increase access opportunities for broad user communities
Cancer Research Services (canSERV)		Europe	Bio-health	Top-down (EC call) + bottom-up multidisciplinary consortium of 19 European partners in the field of oncology/2022 (3y project)	Provides cutting-edge customised research services to the cancer research community	Focus on a common major objective
Coalition Publica	Tanja Niemann	Canada	Social sciences and Humanities	Bottom-up/2017 (currently under a 6y project grant since 2023)	Partner to the European OPERAS initiative; non-commercial, open source national infrastructure for digital scholarly publishing, dissemination, and research	Increase access opportunities for broad user communities
Data from PHoton and Neutron Experiments for the NFDI (DAPHNE4NFDI)		Germany	Physics RI data	Top-down + bottom-up/2021 (5y project grant initially)	Part of a nation-wide effort (National Research Data Infrastructure (NFDI)) to systematically index, edit, interconnect and make available the valuable stock of data from science and research.	Increase access opportunities for broad user communities
Earth Biogenome Project (EBP)	Paul Hebert	International	Biodiversity	Bottom-up/2022 (still in phase 1 development)	Has a strong focus on standardisation of genomic data	Focus on a common major objective
European Intergovernmental		Europe	Fundamental research in natural and physical	Bottom-up/2002	Partnership between European intergovernmental research organisations based on a jointly signed charter	Lobbying towards policy bodies and

Research Organisation forum (EIROForum)			sciences			exchange of information on issues of common interest
Engineering Research Infrastructures for European Synergies (ERIES)	Gerard O'Reilly	Europe + Canada	Structural, seismic, wind and geotechnical engineering	Bottom-up/2022 (4y project)	Focus on risk assessment and risk mitigation	Focus on a common major objective
EUROfusion	Tony Donné	Europe	Fusion energy	Top-down (EC Euratom programme) + bottom-up/2014	Main goal is to pave the way for fusion power reactors (very practical objective)	Focus on a common major objective
EUROpean Laboratories for Accelerator Based Sciences (EURO-LABS)	Paolo Giacomelli	Europe	Nuclear and High Energy Physics	Bottom-up/2022 (4y project)	Brings together, the three communities engaged in Nuclear Physics, Accelerator and Detector technology for High Energy Physics.	Share good practice, optimise resources among similar RIs
EuroNanoLab	Vittorio Morandi	Europe	Nanofabrication facilities	Bottom-up/2017	As a consortium of open-access academic cleanrooms, EuroNanoLab works in complementary with other cleanroom networks around the world (cooperation with US (NNCI), Australia and Japan networks)	Share good practice, optimise resources among similar RIs
EUROPLANET2024 RI	Nigel Manson	Europe + international	Astronomy (planetary sciences)	Bottom-up (follow up to previous programmes supported by the EU/2020)	Provides transnational access to world-leading laboratory facilities, Virtual Access to state-of-the-art data services and tools, and networking activities with partnerships with underrepresented states, non-EU countries and industry	Increase access opportunities for broad user communities
Extreme Light in Intensity, Time, and Space Network (X-lites)		US-International	Laser facilities	Bottom-up/2022	Strong focus on helping scientists (particularly young scientists) to apply to extreme light facilities access and use	Increase access opportunities for broad user communities
German-Canadian Materials Acceleration Centre (GC-MAC)		Canada+Germany	Material sciences	Top-down (policy to support the growth of materials acceleration platforms)/2021 (5 years +)	Support to UN sustainability goals, aligns its R&D strategy with the global transition towards a de-fossilised, decentralised, efficient, and economically viable energy infrastructure	Share good practice, optimise resources among similar RIs + Focus on common major objectives
Global Ecosystem Research Infrastructure (GERI)	Hank Loescher	International	Environmental sciences	Bottom-up/2020 (5y MoU renewed automatically)	Brings together a series of national ecosystem RI networks	Broaden data collection and collaboration at a global scale
Grenoble Innovation for	Francine	Local (Grenoble)	Interdisciplinary	Bottom-up/2008	Focuses on 5 societal challenges and 3 transverse axes.	Increase the

38 | FOSTERING RESEARCH INFRASTRUCTURE ECOSYSTEMS

Advanced New Technologies (GIANT)	Papillon	area in France)			Regroups large basic research RIs, innovation labs from large French research organisations and local higher education institutions	attractiveness of an area
High Value Manufacturing Catapult (HVM Catapult)		United Kingdom	Technology development (Research and Technology Organisations)	Top-down (initiative from the UK Government's innovation agency Innovate UK)/2011	Set up to support the future of manufacturing in the UK, it is part of the wider Catapult Network, where we help to co-ordinate the broader innovation landscape in the UK	Increase competitiveness (innovation focus)
International Barcode of Life (IBOL)	Paul Hebert	International	Environmental observation-biodiversity/DNA sequences	Bottom-up/2010	Strong focus on sharing DNA sequencing technologies	Focus on a major objective
INTERnational Cooperation of high containment research infrastructures: from Epidemic Preparedness TO Response (INTERCEPTOR)	Diana Stepanyan	International	Bio-health (High containment laboratories)	Bottom-up with top-down support (EC grant)/2024 (3y project)	In response to the lessons learned during the COVID-19 pandemic, creates a large pool of capabilities and human resources that can be involved rapidly in urgent high-consequence pathogen outbreak research and response actions	Increase access opportunities for broad user communities + Focus on a major objective
Integrated Services for Infectious Disease Outbreak Research (ISIDORE)	Diana Stepanyan	Europe	Bio-health	Bottom-up with top-down support (EC grant)/2022 (3y project)	In response to the lessons learned during the COVID-19 pandemic, provides streamlined transnational access to RI facilities and services	Increase access opportunities for broad user communities + Focus on a major objective
Japan complementary use programme	Tetsuya Ishikawa	Japan	Quantum beam RIs and High-Performance Supercomputers	Bottom-up + top-down support/2017	Designed to enhance the use of large-scale quantum beam RIs in Japan (synchrotron radiation, X-ray free electron laser and neutron facilities) with the use of the supercomputer.	Increase impact of large RIs
Korea Core Facility Council	Yong Joo Kim	Korea	Multidisciplinary (core facilities from universities)	Top-down (NFEC, that supports the core facility council, is an agency of the Korean government)	The core Facility's primary goal is to create an ecosystem of small-scale RI with a shift towards a more professional operation instead of directly distributing it to individual researchers. Another goal is to foster medium-sized RIs and establish a linkage system with Korea's current large-scale RIs and international RIs to create a pan-national RI ecosystem.	Increase access opportunities for broad user communities
Korea infrastructure Organisation for Nano-technology (KION)	Gyeongyeo p Lee	Korea	Nanotechnologies	Bottom-up with top-down support/2011	KION was formed to maximise the utilisation of individual RIs by exchanging manpower among nano research infrastructures, creating/operating joint utilisation	Share good practice, optimise resources among

					platforms, and fostering shared human resources	similar RIs + Increase access opportunities
Laserlab Europe	John Collier	Europe	Laser RIs	Bottom-up/2004 (opportunity through the EC Framework programme funding mechanism)	Instrumental in increasing role of laser facilities in European science and in the creation of the European Light Infrastructure (ELI)	Share good practice, optimise resources among similar RIs + Increase access opportunities
LaserNet US	Chandra Curry	US + Canada	Laser RIs	Top-down (DoE funding programme)/2018	Strong focus towards opening up facilities to young researchers (several specific schemes)	Share good practice, optimise resources among similar RIs + Increase access opportunities
League of European Accelerator-based Photon Sources (LEAPS)	Jean Dailant	Europe	Synchrotron Radiation and Free Electron Laser user facilities	Bottom-up/2017 (now in the process of becoming a legal entity (LEAPS-AISBL))	LEAPS has enabled many technological developments that benefited to all its member, including best practices shared to face the strong increase of energy price and the reduction of carbon footprint	Share good practice, optimise resources among similar RIs
League of advanced European Neutron Sources (LENS)		Europe	Neutron facilities	Bottom-up/2018	Critical role in advocating for sustainable neutron infrastructure in Europe and is involved in interdisciplinary networks involving material sciences	Share good practice, optimise resources among similar RIs
National Crystallography Service (NCS)		UK	X-ray diffraction facilities	Bottom-up with top-down support/2022 (5y grant)	Provides a national single crystal X-ray diffraction service to tackle problems that individuals cannot address with their local facilities and an access route to synchrotron-based experiments	Increase access opportunities for broad user communities
National Epitaxy Facility (NEF)		UK	Semiconductors facilities	Bottom-up with top-down support/2022 (5y grant)	Strong impact assessment strategy	Increase access opportunities for broad user communities + Increase competitiveness
National Wind Tunnel Facility (NWTF)		UK	Wind tunnels	Bottom-up with top-down support /2014	Builds an Experimental Database to catalogue open access research outputs and promote better connectivity with the computational fluid dynamics community	Increase access opportunities for broad user communities
Ocean Facilities Exchange Group (OFEG)		Europe	Research vessels	Top-down/First barter agreement signed	Unique network that operates on a barter system, enabling joint cruises, ship-time exchange, and	Share good practice, optimise

40 | FOSTERING RESEARCH INFRASTRUCTURE ECOSYSTEMS

				between France, Germany and the UK in 1994, OCEG-TECH set up in 2007	equipment sharing without the exchange of money. Since 2007, OFEG-TECH proposes extended services through the interoperability of large equipment, adoption of common working practices, harmonised mechanisms, protocols and tools.	resources among similar RIs
Ocean Network Canada	Benoit Pirene and Kate Moran	Canada	Environmental observation of the oceans	Bottom-up/2007	Strong links with other international ocean observation networks such as OOI	Broaden data collection and collaboration at a regional scale
Ocean Observatories Initiative (OOI)	James Edson and Paul Matthias	US + international	Environmental observation of the oceans	Bottom-up/2016	Changed ocean research by democratising access to ocean data.	Broaden data collection and collaboration at a global scale
Open Data Infrastructure for Social Science and Economic Innovations (ODISSEI)	Tom Emery	Netherlands	Social sciences	Bottom-up/2016 (strong support from memberships)	Provides a unique and secured access to data from the National Statistics agency	Increase access opportunities for broad user communities
Open Scholarly Communication in the European Area For Scholarly Communication (OPERAS)	Pierre Mounier, Suzanne Dumouchel and Yannick Legre	Europe + Brazil, Canada, Georgia	Social sciences and humanities	Bottom-up/2020 (5y grant from 2022)	Coordinate and federate resources in Europe to efficiently address the scholarly communication needs researchers in the field of SSH. Accepts public and private members (i.e. publishers)	Increase access opportunities for broad user communities
Particles, Universe, NuClei and Hadrons for the NFDI (PUNCH4NFDI)	Thomas Schoerner-Sadenius	Germany	Particle, astro-, astroparticle, hadron and nuclear physics	Top-down + bottom-up/2021 (5y project grant initially)	Part of a nation-wide effort (National Research Data Infrastructure (NFDI) to systematically index, edit, interconnect and make available the valuable stock of data from science and research. Supports data-intense physics at large facilities.	Increase access opportunities for broad user communities
Prairie biomanufacturing hub (Vaccine and Infectious Disease Organization VIDO)		Canada	Pandemic preparedness	Bottom-up with top-down support/2023 (4y initial project)	Bridges basic inter-disciplinary research, commercialisation and biomanufacturing. Component of Canada's Biomanufacturing and Life Sciences Strategy (see box 7).	Focus on a major objective
RadioNet	Izabela Rottmann, Simon Garrington and Aga Slowikowsk	Europe + international	Radio-Astronomy	Bottom-up/2020 (Consortium)	Develops a road-mapping exercise for European radio astronomy, collaborating with European funding agencies, further developing close connections with radio astronomy partners outside Europe and with optical astronomy and astroparticle RIs	Share good practice, optimise resources among similar RIs

	a					
REcyclable MAterials DEvelopment at Analytical Research Infrastructures (REMADE-ARI)	Barbara Schramm	Europe	Material sciences/interdisciplinary	Bottom-up with top-down support/2022 (4y project)	Consortium with a strong interdisciplinary dimension, the project includes a strong activity on impact assessment	Focus on a common major objective
South African Environmental Observation Network (SAEON)	Mary-Jane Bopape	South Africa	Environmental observation	Bottom-up with top-down support/2002	Strong local integration of distributed nodes	Increase access opportunities for broad user communities
UK National Ion Beam Centre (UKNIBC)		UK + international link	Ion beam facilities	Bottom-up with top-down support/2022 (5y grant)	Provides specialist ion beam processing and ion beam analysis facilities to both academic and industrial clients within the UK and beyond	Increase access opportunities for broad user communities
Worldwide LHC Computing Grid (WLCG)	Simone Campana	International	Global computing facilities	Bottom-up/2002	Provides global computing resources to store, distribute and analyse the data generated by the four experiments ALICE, ATLAS, CMS and LHCb at the Large Hadron Collider (LHC), making the data equally available to all partners, regardless of their physical location.	Increase access opportunities for broad user communities
X-Lites	Theodore Ronnigen	International	Laser facilities	Bottom-up/2022	X-lites is a network of networks.	Share good practice, optimise resources among similar RIs and broaden user communities

Annex E. RI ecosystem good practices examples

In this annex, some generic information (on the objectives, organisation and achievements) of a number of RI ecosystems case studies are provided, together with additional details that illustrate some interesting and good practices (in shaded cells).

3. Australian National Collaborative Research Infrastructure Strategy (NCRIS) Health Group

Objectives	To advocate for, and to promote cooperation, collaboration, communication and reciprocal facilitation specifically between the members, and also with other related NCRIS supported capabilities.
Rationale	There are benefits to Health Group members from participation in the ecosystem. These include sharing of knowledge and expertise, joint appointments and coordinated communication channels. Members have also worked together on joint funding submissions. There are also benefits to RI users who may experience more coordinated access to a range of national RIs and have greater success with research grant proposals through leveraging the ecosystem. The breadth of exposure to the health and medical research system through the five participants far exceeds what each organisation could achieve as a single entity.
Organisation	The NCRIS Health Group is a network governed by a MoU.
Achievements	• A better coordinated approach to delivery of health and medical RIs in Australia – including demonstrable examples across areas such as cancer, traumatic brain injury and cardiovascular disease. Greater mutual awareness of capabilities and what the NCRIS Health Group can offer its stakeholders. • A proactive culture within the NCRIS Health Group of seeking out opportunities for collaboration. • Joint staff appointments and initiatives between NCRIS Health Group members. • Stronger connections between Australian national RIs and EU counterparts. • Shared capacity to promote NCRIS supported capabilities to the broader research community (locally and internationally), specifically to industry. • Building the capacity to have a shared and thus a more powerful voice in terms of advocacy and for issues the RI community faces. • Shared advocacy with Australian health and medical research systems (NHMRC, Medical Research Future Fund) in delivering research infrastructure at a local and national scale.

4. League of European Accelerator-based Photon Sources (LEAPS)

Objectives	• Encourage and facilitate discussion and exchange among its members on issues relevant to the shaping of future science and technology at accelerator-based light sources in a European perspective; • Develop common objectives in line with the objectives of its stakeholders to the benefit of European science and society; • Increase coordination between members; • Develop and periodically update landscape documents, roadmaps and action plans for key technologies and promote a collective strategy across European facilities; • Develop common objectives in line with the objectives of its stakeholders to the benefit of European science and society.
Organisation	Launched in 2017 as a consortium in the process of becoming a legal entity (LEAPS-AISBL)
Achievements	LEAPS is key for the use of resources by avoiding the duplication of efforts and for maximising the scientific, economic and societal impact. LEAPS has enabled many technological developments that benefited to all its member. LEAPS has also a strong visibility towards the European Commission and constructive lobbying actions have been successfully conducted (to be further expanded). Tangible results have been achieved, whether in jointly seeking European funding, sharing best practices, or launching large-scale operations in technology developments that could not have been achieved individually. LEAPS has also been instrumental in addressing recent crises affecting the community, such as: <i>Covid 19 Pandemic:</i> All the infrastructures have been impacted by the shutdown. LEAPS helped to share best practices in a timely manner. Much has been learned in terms of optimisation of remote access. A strong

	coordinated initiative was also the opening of strategic access during the Pandemic for projects related to the Covid 19. <i>Energy Crisis:</i> best practices are shared to face the strong increase of energy, the reduction of carbon footprint, e.g., the development of permanent magnets. <i>“Light for Ukraine”:</i> LEAPS designated SOLARIS National Synchrotron Radiation Centre (Poland) as a coordinating facility to implement and supervise the activities aiming at consolidation and strengthening the Ukrainian users’ community. Within the “Light for Ukraine” project the Polish and Ukrainian synchrotron radiation user communities, acting together, are making efforts to build a joint research infrastructure that will further strengthen cross-border cooperation. <i>Africa:</i> LEAPS is being very active to have a call launched by the EC (call Infra Serv) to help structuring the community and networks in Africa in light sources.
--	---

5. Japan Advanced Research Facilities Platform program (ARFP)

Objectives	- Establishment of one-stop service centred on coordinating organisation, mutual cooperation for the use of each organisation’s facilities - Sharing of knowhow and data for remote use and automation of experiments - Assignment, training of professional staff
Organisation	Network managed by a committee set up by Japan’s Ministry of Education, Culture, Sports, Science and Technology (MEXT) and steering committees for each discipline (4 discipline-based platforms), but still a loose organisation with no legal structure; ongoing discussion for a more sustainable structure and regarding the degree of integration
Achievements	Increased visibility towards potential users, exchange of know-how between facilities, some exchanges of resources, and increased support for users to help determine the best facility suitable for the project
Geopolitics	The ARFP was established in 2021 for 5 years initially; the main driver was the Covid crisis that impeded the use of many facilities. From 2016-2020, the MEXT had been promoting cross-institutional efforts to form platforms for innovation by supporting several institutions with advanced research facilities that wanted to collaborate and by requiring a central contact point and an advanced use support system for their use within each platform. However, COVID-19 resulted in restricting the mobility of researchers, making access to research facilities difficult. In particular, the limited availability of other alternatives for some of the leading advanced research facilities had a significant impact on research activities, with the entire research plan being reviewed and/or delayed. In response, MEXT decided to provide support to remote use and automation to maintain research activities. The objective was also to improve the efficiency of facility use and the capacity to conduct experiments remotely, which in turn would promote improvements in the use of advanced research facilities. The initial goal of ARFP was thus to promote the use of the facilities and share user access practices (particularly the development of remote access and data sharing). Following an open call by the MEXT, 4 networks were selected for support in the disciplines of NMR, microscopic imaging, power lasers and research MRI, and activities to improve user and data access are underway in each network.

6. German Canadian Materials Acceleration Centre (GCMAC)

Objectives	Coordinate and integrate German and Canadian activities in accelerated materials research: - Support technology-focused materials research. - Advance method and tool development. - Accelerate discovery, design, and scale-up of energy materials. - Provide education and training programs - Knowledge sharing & network/ecosystem building
Organisation	Collaborative partnership of research activities between Germany and Canada, funded by matching (mostly in-kind) resources from participating institutions, linked by an Agreement for Cooperation (April 2021). Includes a virtual office supported by Germany and Canada which provides administration support, secretariat, communication, and project management support, and management committee consisting of the principal investigators from Germany and Canada.
Achievements	GCMAC deployed one of the most highly interdisciplinary project scopes of any major project but with a modest budget allocated to activities like collaborative support actions rather than flagship projects. It aligned facilities, capabilities and expertise in two countries in a common goal to accelerate materials discovery in the important topic of hydrogen production as an energy carrier.
Geopolitics	The idea for the creation of GCMAC evolved from activities under the umbrella of the global initiative Mission Innovation (MI), Clean Energy Materials Innovation Challenge IC6 (2017-2021), later Materials for Energy (M4E) Collaborate Platform (2021-present), currently co-led by Germany and Canada. Concurrently, the ecosystem has grown through engagement with other related international initiatives, with a focus on joint activities to promote knowledge sharing and network building.

7. LaserLab EU

Objectives	- To maintain a competitive, inter-disciplinary network of European national laser laboratories; - To strengthen the European leading role in laser research through Joint Research Activities (JRA), pushing the laser concept into new directions and opening up new applications of key importance in research and innovation; - To offer transnational access to top-quality laser research facilities in a highly co-ordinated fashion for the benefit of the European research community; - To increase the European basis in laser research and applications by reaching out to neighbouring scientific communities and by assisting in the development of Laser Research Infrastructures on both the national and the European level
Organisation	Initially a cooperative undertaking between European laboratories, evolved into an Integrated Infrastructure Initiative (I3) supported by the European FP6 programme, and currently in transition to a more permanent organisation.
Achievements	LaserLab fostered the development of an internationally recognised structured community, very visible in Europe and highlighting the European laser science. The creation of ELI (European Light Infrastructure) stemmed from discussion and lobbying from LaserLab. LaserLab was instrumental in increasing the role of laser facilities in European science, for example the 2023 Nobel prize in Physics was linked to experiments conducted using lasers (Pierre Agostini, Ferenc Krausz and Anne L'Huillier received the award for having developed methods that produce ultrafast laser pulses, which have applications in chemistry, biology and physics).
Governance	The governance structure for LaserLab as a I3 project is fairly standard with a general assembly, and administration office (https://www.laserlab-europe.eu/about-us/governance). When information about the end of the I3 came in, two possible models were discussed: an association under Belgium law (AISBL) or a European consortium (ERIC). The choice for an AISBL was determined because of its simplicity and because it can accommodate institutions as members: only institutional agreements were required, not country agreements. This has proven to be efficient in managing the network and useful when trying to approach or influence EC projects/calls. The role of LaserLab as an AISBL is that of a coordinator and manager of the network, but it does not have the capacity to conduct joint research projects; this is a problem because many members are small RIs that are not access providers but just research facilities that are mostly interested in developing joint R&D projects. Hence there are discussion whether the ERIC solution might still be adopted. This would provide more core funding and hence allow the development of joint research activities.

8. Ocean Facilities Exchange Group (OFEG)

Objectives	- To represent Europe's leading oceanographic research organisations - To provide a forum to consider barter exchange and co-operation opportunities for the Global and Ocean Class research fleet. - To maximise the overall scientific output using its state-of-the-art facilities in support of the worldwide oceanographic community - To develop interoperability of large equipment, the adoption of common working practices, harmonised mechanisms, protocols and tools
Organisation	OFEG functions as a collaborative partnership fostering information sharing and resource exchange. Originally a tripartite agreement between Germany, France and UK; since joined by the Netherlands, Spain and Norway. OFEG is an equal partnership between all members. OFEG-TECH, a more recently established subsidiary of OFEG, was founded in 2007 to aid the development of OFEG from a pure ship barter organisation to one that is also capable of delivering state of the art marine scientific facilities across all OFEG platforms, through interoperability of large equipment, adoption of common working practices, harmonised mechanisms, protocols and tools.
Achievements	Compared to working as individual entities, the OFEG ecosystem provides scientists with significantly wider access to research facilities and equipment; OFEG boasts a combined fleet of 21 research vessels, including a majority of Europe's "heavy" equipment like submersibles, ROVs, and advanced survey systems, which individual institutions might otherwise find too expensive to acquire on their own. This access extends beyond just quantity, as OFEG includes 10 out of 11 Global Class and 8 out of 15 Ocean Class research vessels in Europe, offering access to cutting-edge technology. Additionally, OFEG fosters cost-effectiveness by optimising fleet movements based on the geographical location of ships, reducing travel time and associated costs. This allows scientists to access a broader range of geographical regions for research within a single year.
Business model	OFEG operates on a barter system, enabling joint cruises, ship-time exchange and equipment sharing without the exchange of money. A points value is agreed for each asset and a running total or tally is kept for all partners based on how much they take advantage barterers as much as offer them. There have been a small number of instances of charter within the OFEG group, but these have also come with added value as they have been run

	at cost hire rather than the market rate. There is no drive behind OFEG to operate on a profit basis. All partners commit to the running of an OFEG website that centralises requests and offers and there are small monetary contributions for all partners to keep this running.
--	--

9. Engineering Research Infrastructures for European Synergies (ERIES)

Objectives	To provide transnational access to advanced research infrastructures in the fields of structural, seismic, wind and geotechnical engineering.
Organisation	ERIES is a project consortium (12 RIs from 7 European countries + Canada) supported by EU Horizon Europe programme (running project from 2022 to 2026).
Achievements	ERIES has broadened past networks focused on earthquake engineering to other hazards, gathering complementary facilities for more interdisciplinary activities. It is a RI ecosystem aiming at impact beyond research i.e. civil authorities, risks mitigation and risks management etc.
User and data access	ERIES operates a unique user access mechanism, which is dedicated to transnational access (access of the facilities to users outside the facilities' host country). After an initial interaction with the targeted facilities, user groups submit their proposals via a Google form to a centralised Transnational Access Selection and Evaluation Panel (TA-SEP) which is in charge of reviewing the applications and selecting the projects to be funded and the hosting facilities. Each research infrastructure provides feedback to the TA-SEP for the proposals submitted requesting access to their installations. Competing proposals for each research infrastructure's installation are ranked based on the total score they receive from the TA-SEP. Experimental data produced at ERIES facilities are stored and organised via online tools following FAIR principles. All metadata and other information are available on a Data Access Portal.

10. ODISSEI

Objectives	To improve user access to social sciences and humanities (SSHs) RIs. ODISSEI provides analytical tools, algorithms, secure computing environment and specialised services and expertise.
Organisation	ODISSEI is collaborative consortium of 45 member organisations in the Netherlands. It is financed structurally by its members for up to €1 million per year (the total Operational Budget is ~€5 million per year).
Achievements	ODISSEI coordinates all the Social Science Infrastructure of the Netherlands in a unique consortium. Social Science is traditionally a very scattered field with very different data access policies and data standards. Not only has ODISSEI been able to coordinate a wide range of social research institutions in the Netherlands but also includes and provides secured access to data from the National Statistics agency (Centraal Bureau voor de Statistiek). Secure access to the large diversity of social science data has allowed the conduct of complex interdisciplinary studies of societal interest (such as the creation of population scale networks and high-resolution digital twins).
Data access	ODISSEI is unique in enabling longitudinal scientific studies to be linked to complex and high-dimensional administrative data in a secure HPC environment. This includes a rich a diverse range of data including administrative data, digital trace data, biomedical data, and high-quality longitudinal survey data. Under the CBS Wet and GDPR, these data can be linked at the individual level and accessed by researchers under secure but practicable conditions. ODISSEI enables access under the five safes principles via verified and audited secure environments and supports open science and community development under these secure conditions via bespoke and specifically developed tools. As the national statistics institution and government research agencies were brought into the consortium, this also allowed the development of interoperability standards between research and government data sets.

11. LaserNet US

Objectives	• Advancing the frontiers of laser-science research; • Providing students and scientists with broad access to unique facilities and enabling technologies; • Fostering collaboration among researchers and networks from around the world.
Organisation	LaserNetUS is a consortium of premier high-power laser facilities across North America, including Canada. Established by the U.S. Department of Energy's Office of Fusion Energy Sciences (DOE-FES) in August 2018, the network comprises national laboratories and top-tier universities renowned for their unique and complementary laser and technical capabilities. By bridging academia and private industry, LaserNetUS supports a wide range of interdisciplinary research and development initiatives. Managed by SLAC National Accelerator Laboratory, LaserNetUS is led by the Coordinator and supported by the administrative staff of the LaserNetUS Users Office. Their role is to oversee and execute the proposal review process, research grants, and user initiatives.
Achievements	LaserNetUS has provided invaluable support for industry-relevant experiments through beam time allocations and expert assistance. The network has established strong connections with key fields such as plasma science, material science, optics, laser science and technology, medical applications, advanced manufacturing, and

	national security. The organisation has also established outreach programs, including a mini-interview series and blogs on popular science topics related to high-power lasers, increasing exposure to and understanding of ongoing research. Additionally, the newly established Technical Exchange Program optimises operations across facilities through coordination, robust communication, and in-person visits, fostering continuous improvement and collaboration across the network.
Training	LaserNetUS has developed a robust training program tailored to its user community, especially young scientists. Through initiatives like Research Proposal Support, it provides guidance to maximise the success of research proposals and offers hands-on experience with state-of-the-art laser systems. The Student Ambassador Program enables graduate students to enhance their presentation skills, expand their professional networks, and gain critical exposure for their future careers. These initiatives have led to numerous Ph.D. theses, invited presentations at major conferences, and the successful placement of students in post-graduate jobs at national laboratories and fusion energy companies.

12. UK National Epitaxy Facility (NEF)

Objectives	The National Epitaxy Facility's mission is to provide services for the provision of semiconductor epitaxial materials, structures, and devices to enable the UK's research community to compete at the highest levels on the world-stage in this highly active field. They support UK university research and industry development through the provision of high-quality Semiconductor Epitaxy for custom designed structures and devices.
Organisation	NEF is a consortium between the Universities of Sheffield, Cambridge and University College London and is funded by the Engineering and Physical Sciences Research Council as part of a network of National Research Centres. The NEF has a steering committee comprising of leading academics and industrialists from both the UK and Europe. They meet every 6 months with the Management Team to review the performance of the facility, provide feedback, and advise on improvement opportunities.
Achievements	A broad range of novel devices are produced through research done in the Epitaxy Facility, including LEDs, lasers, quantum circuits, transistors, advanced detectors, and solar cells.
Impact assessment	NEF's impact and success are measured not only through KPIs and risk management but through a diversity of impact criteria. The first are based on scientific achievements and input. They keep an updated list of publications that arise from the work on their materials, especially where their staff has been named as authors on the paper. NEF also tries to demonstrate the impact of their work on a broader societal or economic basis. This is part of making their case for funding to the EPSRC. NEF's impact is also assessed regarding their coordinating role, showing how they are facilitating and supporting the community and the health of that community itself. The NEF also highlights their advocacy and community support activities, such as running conferences, helping with workshops, and being involved in various scientific and advisory committees for other facilities. They also work with institutes for training. These activities are part of their role but are not KPI-related.